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Invited review

Facility location decisions for drone delivery: A literature review

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ABSTRACT

This study presents a comprehensive literature survey on facility location problems for drone (uncrewed vehicle) delivery, where either (i) drones are the only vehicles, or (ii) drones and other vehicles (e.g., trucks) work together for delivery, but drones do not ride in or on the other vehicles. The main goals of this review are to identify and categorize fundamental facility location problems associated with drone delivery, to document the large volume of research in this area, to provide a connection between the studies from different research fields that consider similar location problems, and to highlight promising areas for future research. We first discuss and classify the functions of drones and the various types of facilities used for drone and hybrid vehicle-drone (e.g., truck and drone, or transit and drone) delivery systems, including drone bases (fixed or temporary), other vehicle bases, recharging stations, (re)supply points, and platooning points. The literature is reviewed and categorized based on the types of vehicles involved and their interactions, the types of facilities located, the types of drones and the location space (discrete or continuous). Each category is analyzed in terms of the modeling approach, decision(s), the objective function(s), constraint(s) and additional feature(s). The paper concludes with some promising future research directions.

1. Introduction

There is a rapidly growing body of literature for drone delivery problems, which often involve facility location decisions. A delivery drone is an uncrewed vehicle that can carry an item to be delivered, and besides aerial drones, this can include ground drones, as well as sea surface drones and undersea drones. Different types of facilities may be utilized in drone delivery systems, so this article first presents a discussion of the functions of delivery drones and their associated facilities. We then provide a classification and discussion of the relevant literature. In this review, we focus on drone delivery research where (i) drones are the only vehicles, and (ii) drones and other vehicles (e.g., trucks) work together for delivery, but drones do not ride in or on the other vehicles. The companion paper (Dukkanci, Campbell, & Kara, 2023) provides a review of the studies with facility location decisions where drones ride in/on other vehicles. For research on routing aspects of drone delivery with riding, see Chung, Sah, and Lee (2020), Li, Chen, Wang, and Bai (2021), Macrina, Pugliese, Guerriero, and Laporte (2020) and Moshref-Javadi and Winkenbach (2021).

This literature review is based on an analysis of journals and online publications from a wide range of fields including operations research, management science, engineering, transportation, public health, geography, computer sciences, robotics etc. The research publications

use a variety of terms for the autonomous vehicles, including drone, autonomous vehicle, UAV, unmanned aerial vehicle, uncrewed aerial vehicle, UGV, unmanned ground vehicle, and robot; but we will use the term *drone* in this review. Fig. 1 shows the number of articles per year from 2016 to 2023, where the number of papers for 2023 is for only part of the year.

This literature review does not cover studies that consider the location of drone-related facilities for telecommunications and sensing, monitoring or surveillance applications. For telecommunication applications, see Boccadoro, Striccoli, and Grieco (2021), Cicek, Gultekin, Tavli, and Yanikomeroglu (2019), Elnabty, Fahmy, and Kafafy (2022), Mozaffari, Saad, Bennis, Nam, and Debbah (2019) and Hentati and Fourati (2020). For sensing, monitoring, surveillance or agricultural applications (e.g., spraying crops), see Cabreira, Brisolara, and Paulo R (2019), Mogili and Deepak (2018), Radoglou-Grammatikis, Sarigianidis, Lagkas, and Moscholios (2020) and Rejeb, Abdollahi, Rejeb, and Treiblmaier (2022). (Note though that some drone research on monitoring/surveillance and telecommunications applications can be modeled similar to drone delivery if a set of nodes is specified to be visited by a drone, even if no item is delivered.) This survey also does not review drone delivery research for transporting people or weapons, for sea surface drone applications, or for within-facility applications,

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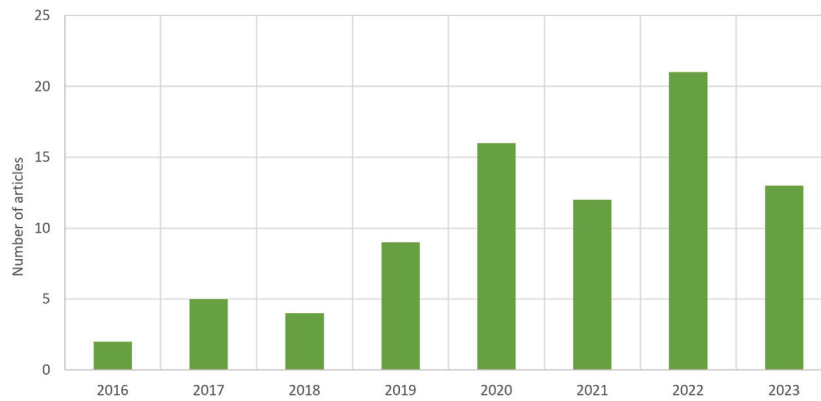


Fig. 1. Number of articles per year.

including automated guided vehicles (AGVs). For these areas, see Gu, Goetz, Guajardo, and Wallace (2021), Fragapane, De Koster, Sgarbossa, and Strandhagen (2021) and Srinivas, Ramachandiran, and Rajendran (2022), respectively. For a broader review on civil applications for drones, see Otto, Agatz, Campbell, Golden, and Pesch (2018), and for a review on charging station location problems for electric cars and trucks, see Kchaou-Boujelben (2021) and Majhi, Ranjitkar, Sheng, Covic, and Wilson (2021).

There are several potential advantages to using drones to make deliveries instead of more conventional delivery options such as crewed ground transportation (e.g., trucks). Drone delivery is usually lower cost than conventional truck delivery due to lower energy (fuel) costs and not paying for driver/delivery personnel, including wages, benefits, insurance, etc. Additionally, drone delivery may provide faster service for customers as drones may have a higher speed than trucks, travel a shorter distance than the road network, and can avoid road traffic congestion. Fast service is crucial for many healthcare delivery applications, as medical items such as blood, defibrillators, and some vaccines are time-sensitive. Drone delivery provides another advantage for humanitarian applications, in particular post-disaster applications. As drones do not require ground infrastructure for travel or delivery, they allow deliveries of critical supplies in regions with damaged and destroyed ground infrastructure (e.g., missing bridges, unpassable roadways, etc.). Finally, drones may provide environmental benefits, especially battery-powered drones when compared to conventional diesel trucks, depending on the source of electricity and lifecycle emissions (Baldissieri, Siragusa, Seghezzi, Mangiaracina, & Tumino, 2022; Stolaroff et al., 2018). However, a fair evaluation of the environmental impacts of drone delivery requires comparison with optimized delivery by alternative fuel vehicles.

Along with the potential benefits from drone delivery, there are a number of associated social, legal and policy issues to consider. Key concerns in these areas have been identified as safety and security, privacy, and regulation/legal, including integration with current airspace users (Nisingizwe et al., 2022; Sabino et al., 2022; Smith, Dickinson, Marsden, Cherrett, Oakey, & Grote, 2022) and we agree with (Roberts et al., 2023) that “significant research remains before drones are likely to realize their full potential and achieve widespread adoption, including further research demonstrating functionality in real-world scenarios”, as well as adequately addressing social, legal and policy concerns.

The contributions of this paper are as follows: (i) we provide a classification of facility functions and types for drone delivery, (ii) we highlight major differences between location problems with conventional vehicles and drones, (iii) we provide a general methodological framework for facility location problems for drone delivery, (iv) we document the research intensity for locating different types of facilities in drone delivery systems, (v) we categorize the literature on facility location for drone delivery systems, (vi) we connect research

from OR with research from other disciplines and provide a unifying framework for facility location in drone delivery research, and (vii) we identify relatively unstudied problems that are promising areas for future research.

The remainder of this paper is organized as follows. Section 2 reviews drone functions, facilities, and delivery problem settings, highlights differences between classical delivery and drone delivery and presents a literature categorization scheme. Section 3 provides a general methodological framework. Section 4 reviews the literature for drone delivery facility location problems where drones are the only vehicles. Sections 5–7 review the literature for drone delivery facility location problems where there are drones and other vehicles. Section 5 addresses research where drones and other vehicles have no interactions, Section 6 addresses research where other vehicles (re)supply drones with packages, and Section 7 reviews studies where drones (re)supply other vehicles with packages. Section 8 reviews the literature with novel use of drones that do not fit into other categories. Section 9 presents a summary and discusses future research directions, and Section 10 provides conclusions.

2. Drone delivery settings

This section defines drone functions and drone delivery facilities, and provides our categorization scheme for the literature. The items to be delivered, which can be carried by a drone, are called *packages*. In a delivery problem, demand for a delivery occurs at points (nodes) that we call *customers*. Packages typically originate from a drone depot or a depot of another transport mode (e.g., a truck depot), and delivery requires transporting packages from the depot(s) to the customers. Some research considers pickup problems where items are collected from customers for transport to a depot, and this is generally modeled similar to a delivery problem, but with travel in the reverse direction. There is also some research on problems with mixed pickups and deliveries, where packages originate or terminate at depots, as well as research on pickup-delivery problems where each package can have a specified origin and destination node (e.g., Dhote & Limbourg, 2020; Pinto, Zambetti, Lagorio, & Pirola, 2020). Our general presentation reflects the majority of the literature and focuses on delivery problems from depots to customers.

Delivery drones have a payload limit (how much they can carry) and a range limit (how far they can go in distance or time before needing to be recharged/refueled). The payload limit is actually in dimensions of both weight (kg) and volume (liters), though research rarely considers both limits. In general, drones may transport themselves (e.g., flying) or they can be carried on (or in) another vehicle (e.g., riding on a truck) as in the companion review (Dukkanci et al., 2023). In this review, drones do not ride on another vehicle, so they have the following capabilities: (i) departure from a place, (e.g., a takeoff or launch), (ii) traveling between two places carrying zero packages (i.e., empty) or

Table 1
Functions of drones.

Function	ID	Description
Load	L	transfer package(s) to a drone
Unload	U	transfer package(s) from a drone to a customer, facility or other vehicle
Departure - Flying	DF	initiate a drone flight where the drone moves independently (e.g., takeoff or launch)
Arrival - Flying	AF	terminate a drone flight (e.g., landing, recovery or retrieval)
Charge	C	reset the range of the drone (e.g., charge the battery or swap it with a new battery)
Meet	M	drone meets another drone (e.g., to form a platoon or to exchange packages)
Waiting Hover	WH	drone waits for some time while “active” (could occur in the air or on the ground)
Waiting Park	WP	drone waits for some time while inactive (e.g., parked on the ground and turned off)
Storage	S	drone stays at a facility overnight (or at the end of the time period)
Maintenance (Fix)	F	have a drone repaired or maintained

Table 2
Common facility types and functions for drones.

	L	U	DF	AF	C	M	WH	WP	DR	AR	S	F
Drone Base (DB)	x	x	x	?							x	x
Customer (CP)		x	x	x								
Recharging Station (RS)			x	x	x							
Holding Point (HP)			x	x			x	x				
Platooning Point (PP)		x	x			x	?					
Other Vehicle Base (OB)									?	?		
Temporary Drone Base (DB-T)	x		x	x	?				?	?		
Resupply Point (SP)		x	x	x								

carrying one or more packages, (iii) arrival at a place (e.g., a landing or recovery), (iv) staying in one location while “active” (e.g., hovering or idling), and (v) staying in one location while “inactive” (e.g., parked and turned off). A special case of travel (item (ii)) is platooning, which is when more than one drone travel together, and may be for purposes of efficiency or traffic control.

Within the drone delivery research literature, the facilities used by drones have been given a wide range of different, and sometimes unique, names, so facilities that perform the same functions appear in different articles with completely different names. For examples, drone station, drone depot, drone base and drone beehive may refer to the same type of facility. To provide a common framework for reviewing this literature, we first describe a comprehensive set of functions performed by various drone facilities, which then allow us to coherently define different facility types and classify the literature. Note that drone facilities may be stationary (e.g., a building) or mobile (e.g., a truck). The fundamental functions of drone facilities along with their abbreviations and descriptions are given in Table 1. Each of the functions has a one or two-letter identifier shown in the second column of Table 1. In general a facility performs several of these functions; for example, in a simple drone delivery problem with one drone depot and no other vehicles, the drone depot (or base) includes L, DF, AF, C, S, and F.

We define a *drone facility* as any location where one or more of the 10 functions occurs. In some cases these could be large buildings, but in other cases they might be locations with no infrastructure, as for a vacant space where a truck parks to launch or retrieve drones, or a point in space where drones meet. Facilities are generally modeled as nodes in a network or points in two-dimensional space. Vehicles begin and end travel from their respective depots (e.g., truck depot, drone depot) and in most of the literature packages are assumed to originate at these depots as well. Every delivery requires a vehicle and package arriving together at a customer. Drone delivery research

generally does not model the inbound transportation of packages to a depot, nor does it usually model inventory at the depots (or elsewhere). Usually, a set of customers (destinations) is given and depots as well as intermediate facilities that perform some of the functions listed above (e.g., recharging) need to be located.

To allow a coherent presentation of the research, we classified the drone delivery facilities in the literature into eight types. Each facility may perform several of the functions listed in Table 1, and a particular location may include several different facilities (e.g., an industrial customer site may provide drone recharging), so we do not define a facility class for all combinations of all ten functions above. Table 2 shows the correspondence of the eight facility types and the ten drone functions in Table 1. The first five facilities (above the line) are for settings with drones (DVs) and no other vehicles. The lower three facilities reflect problems with drones (DVs) and other vehicle types (OVs). The table shows with “x” the typical drone functions for that facility, and with “?” other functions that may occur there. We first describe the facility types below.

In delivery problems with only drones, we identify five facility types. (1) A *drone base* (DB) is a location that provides the following functions: L, U, C, DF, AF, S, F. This is fundamentally the starting point of drone trips, and in a simple delivery setting the package is loaded on the drone at the drone base. (2) A *customer point* (CP) is the location where demand for delivery of a package occurs, so the primary functions are U, AF and DF. (3) A *recharging station* (RS) is a location that provides functions C, AF and DF. (4) A *holding point* (HP) is a location that allows the drone to wait before proceeding with a subsequent action (e.g., a drone may hover at an HP to wait for a recharging station to become unoccupied), so the primary functions are AF, DF, and WH and/or WP. (5) A *platoon point* (PP) is a location where two or more drones meet to begin a platoon trip (i.e. two or more drones traveling together) or where they separate at the end of a platoon trip, so the primary functions are AF, DF, and M. Note that CPs can be either given locations or determined by the optimization, as when an individual may receive a delivery at home, work, or in a locker elsewhere.

When the delivery setting also includes vehicles other than drones, such as trucks or transit vehicles, then additional types of facilities may be located. For settings with both DVs and other vehicles (OVs), we define three additional facility types. (6) An *other vehicle base* (OB) is fundamentally the starting point of trips for the other vehicles. By definition, an OB location does not have any functions of DVs defined in Table 1 since OBs only serve OVs. However, DBs and OBs may be in the same physical location as for a distribution center that dispatches both trucks and drones. (7) A *temporary drone base* (DB-T) is a location from which drones can depart and arrive, but they cannot stay beyond the end of the day (or time period). It generally provides the following

functions: L, DF, AF. It may also provide a charging function (C). (8) A (re)supply point (SP) is a location where a drone (re)supplies another vehicle (e.g., a truck) with packages that the other vehicle then delivers. This has functions AF, DF, and U.

2.1. Differences from classical facility location problems

This subsection identifies important differences between location problems for drone delivery and classical (non-drone) facility location problems. We focus here on small aerial drones, as they are the most common in the research literature (ground drones obviously have more similarities with trucks). Most drone delivery research uses modeling elements derived from classical delivery problems, and some drone delivery mathematical models are very similar (even identical) to existing facility location or delivery models with trucks (or other ground vehicles). Yet, drones are very different than trucks and other ground vehicles, which suggests that some new modeling approaches may be useful. The important differences between delivery and location problems with drones and with trucks stem from several fundamental differences with drone delivery: (i) drones are autonomous so there are no human drivers or pilots, (ii) the fundamental forces that affect air travel of small craft are different than those for ground (and water) travel of larger vehicles, (iii) drone trips are in three-dimensional space with multiple components (takeoff, landing, ascending, descending, cruising), and (iv) delivery drones have small size and mass.

We briefly discuss the implications of these areas of difference, keeping in mind that many of these are interrelated.

- (i) *No human drivers or pilots*: Removing the human drivers removes or alters the costs and/or constraints associated with people, such as wages, insurance, hours of service regulations, etc. However, drone operations that require operators on the ground would incur some of those costs. Drone delivery operations also have no need to return drivers to their home bases which could allow more flexible delivery routes.
- (ii) *Fundamental forces for flight*: The fundamental forces affecting truck travel are mainly rolling resistance and aerodynamic drag, which are quite different than for an aircraft. The fundamental forces for flight are weight, lift, thrust, and drag, so modeling the cost or energy consumption for drone flights should account for those as appropriate. Most classical OR facility location and delivery models have a single vehicle type and treat transportation rather simply with a constant cost rate (\$/km) or speed throughout the network. In this case, when stopping cost and time are ignored, then minimizing total travel distance also minimizes total cost and completion time (makespan). However, drone travel is more complex and drone delivery with other vehicles involved means that minimizing travel distance (or time) does not necessarily minimize time (or cost). Thus, classic facility location models that assume constant cost rates or speeds may not be the best template for drone delivery models. Energy (fuel) consumption for ground vehicles does depend on factors other than just travel distance, including the payload, speed, roadway gradient, pavement type, congestion, gear selection, etc.; and some research does use nonlinear models to capture some of these factors (e.g., see Demir, Bektaş, & Laporte, 2014; Dukkanci, Kara, & Bektaş, 2019; Khoei, Süral, & Tural, 2017; Mohammedi, Torabi, & Tavakkoli-Moghaddam, 2014). But most drone delivery research, similar to location problems with non-drone vehicles, uses simple distance, time or cost-based objectives without considering the detailed factors affecting drone energy consumption. These factors include the number or size of rotors (if it is rotary-drone), size and weight of the battery, lift-to-drag ratio, avionics, airspeed, motion (takeoff, landing, hover, cruise, and angle and attitude of flight), etc. Zhang, Campbell, Sweeney, and Hupman (2021). There are some examples of DV-based

location problems that use a detailed calculation of drone energy consumption, either in the objective (e.g., Chowdhury, Emelogu, Marufuzzaman, Nurre, & Bian, 2017) or the constraints for drone range (e.g., Bruni, Khodaparasti, & Perboli, 2023; Chauhan, Unnikrishnan, & Figliozzi, 2019; Zhu, Boyles, & Unnikrishnan, 2022).

- (iii) *Drone trips include multiple components in three-dimensional space*: A drone trip requires multiple components: takeoff and climb to a cruising altitude; travel to the destination; delivery of the package, which may require descending, hovering, landing, lowering a tether or releasing a parachute; then a repeat for the return trip, but without the package. Each of these flight components has a unique energy consumption pattern, that can change with each delivery as the payload decreases. The different nature of drone trips can also require different equipment at drone facilities, which affects relevant costs. Classical facility location problems generally do not consider different types of facilities with different functions (e.g., the functions given in Table 1). But the multiple activities of drones may require specialized equipment such as for loading, launching or recharging, which can imply a wide range of fixed and variable costs depending on the specific instance. As shown in Table 2, each drone facility has a different set of functions and each function requires a different amount of space, cost and equipment. For example, a DB might require physical buildings for secure storage, while an RS might need only a small space for a charging platform (Ghelichi, Gentili, & Mirchandani, 2021). Charging/refueling is also an important area of difference between drones and delivery trucks, as there is almost no restriction for diesel and gasoline trucks related to refueling infrastructure. However, drone recharging or battery swapping requires specialized infrastructure, at a smaller scale than for recharging of electric vehicles. For example, Zipline's new home delivery system (Zipline, 2023) uses a DV and a smaller droid carried by the DV. In this system, the DV lands and launches from a charging pad located outside of the DB, while the droid enters the building to be manually loaded with the package. Other systems use small automated drone docking stations for landing, un/loading and recharging, as for Matternet (Coldewey, 2021). Wing proposes a different system, where automated loading and recharging by pads placed on flat surfaces (e.g., roofs) take place in different locations (Wing, 2023). Waiting and hovering are other areas where drones differ from trucks, as drones may hover while making a delivery or waiting for a resource (e.g., a recharging station or landing location) which can consume energy at a rate similar to cruising. Ground vehicle waiting is typically not included in classical location and delivery models (unless there is synchronization), and trucks can idle at low cost or be shut off and secured. In contrast, a waiting drone may need to hover, instead of land, for safety and security reasons, which can have a significant energy consumption, thereby affecting travel distance and the need for recharging. Hovering locations may be restricted to specific locations to reduce airspace congestion and noise, and drones may hover at different altitudes over the same point.
- (iv) *Small size and mass*: The small size of drones means that the range and payload are limited, as the battery mass is necessarily small. Further, a package may be a large portion of the drone mass, so that drone performance before and after a delivery may be different (e.g., the energy consumption rate is lower for a lighter drone after a delivery). This is unlike truck delivery where each package is usually a small fraction of the total vehicle mass. The small battery size of a drone (due to the high cost of flying and heavy weight of batteries) means that recharging may be an attractive option for drone delivery. In contrast, most studies with non-drone vehicles (e.g., trucks) ignore the practical

range limit (e.g., based on fuel capacity), and assume such vehicles have unlimited range. A notable exception is the research on modeling the range of electric vehicles (EVs) and recharging station locations (e.g., Chung & Kwon, 2015; Majhi et al., 2021; Schneider, Stenger, & Goeke, 2014). Because delivery drones have a more limited range compared to EVs, the range limitation is a key feature that needs to be included with drone delivery as a constraint, as part of the data or as a parameter reflecting allowable drone flights. In terms of (re)charging, ground EVs and drones have different charging infrastructure and different types of batteries. For instance, while EVs have both static (plug-in) or dynamic wireless charging options, which might require more space, charging platforms for some drones might be installed on streetlights or billboards (Ghelichi et al., 2021).

Similar to the range, drones have more limited payload capacity compared to OV's such as trucks, so the number of deliveries per route is an important difference between drone and truck delivery. In most DV-based delivery models, DVs make only one delivery per flight, with an assumption that the delivery is less than the DV payload limit. Some research does model multiple deliveries per drone flight (e.g., Bruni & Khodaparasti, 2022; Liu, Deng, Mao, Yang, & Kaisar, 2020; Liu, Yan, Pu, Wang, & Kaisar, 2021; Macias, Angeloudis, & Ochieng, 2020; Torabbeigi, Lim, & Kim, 2020; Uhm & Lee, 2022; Xia et al., 2023) using payload capacity constraints similar to VRP research. As discussed earlier (category (ii)) some non-drone location research considers fuel consumption models that account for vehicle payload, although vehicle range is generally not treated as a constraint. For drone delivery, Bruni and Khodaparasti (2022), Bruni et al. (2023) are examples that consider variable payloads and the impact of payload on energy consumption with multiple delivery drone trips. Finally, the small size of drones means that weather events and wind are likely to be more significant than for larger ground vehicles. The variability and uncertainty of wind speed, wind direction and rain amount directly affect the DV energy consumption, and Kirschstein (2020) is an example showing how such issues can be included for drone delivery.

2.2. Drone delivery examples

This section provides examples to show the use of the eight facility types in drone delivery. The travel links between facilities in a drone delivery problem are of two possible arc types: 1) A *drone arc* (DA) is an arc where a drone can fly; and 2) An *other vehicle arc* (OA) is an arc where OV's can travel. In the illustrations, solid blue lines show loaded drone delivery flights and dashed blue lines show the drone empty return flights. Fig. 2 shows an example for delivery with only drones involving the five facilities in the top of Table 2. Here two drones leave the DB (each with two packages), one is recharged, then each delivers to a customer, and then both meet at a platoon point (PP). The drones travel together to another platoon point, where they separate for deliveries to two other customers. Then, suppose the drones need recharging, but the recharging station can accommodate only one drone at a time. One drone arrives at RS to begin recharging, and the second drone travels to a holding point (HP), where it can safely hover until the first drone finishes recharging. Once the first drone is recharged it returns to the DB and the second drone moves to the RS for recharging; and then once recharged it returns to the DB. Note that some of the facilities in these figures may be co-located at the same node in a model (e.g., for the HP and RS or for the PP and a CP in Fig. 2).

Figs. 3 and 4 provide simple delivery illustrations where drones are used along with another vehicle type. In these figures, the red solid and dashed lines represent loaded and empty other vehicle travel, respectively. Fig. 3 shows the use of temporary drone bases (DB-T). Here a truck that carries two packages departs from the OB and a drone flies from the DB to DB-T1 where the truck supplies the drone. Then,

the drone makes a delivery to a customer and meets with the truck at DB-T2 to be resupplied. After that, the truck returns to the OB and the drone makes the last delivery and flies back to the DB.

Fig. 4 illustrates the use of (re)supply points. Here, a drone flies from DB to SP, where it supplies a package to the OV that traveled from OB, making one delivery on the way. From SP the drone returns to DB, while the OV makes two deliveries before returning to OB. This setting with (re)supply by drone may be advantageous for perishable products (e.g., medical supplies) where the DV travel may be much faster than the OV travel, or in cases where the DV has a large payload and can prevent the OV from having to return to the depot to pick up new packages.

2.3. Categorization

Table 3 shows the number of references from the literature on drone delivery without riding for locating each type of facility. The first column shows the facility type. The next two columns indicate the solution space for the model, the following four columns show the objective, and the next three columns indicate the location of the drone launch and retrieval. "Diff." indicates each drone must be retrieved at a different node from where it was launched. "Same" indicates each drone must be retrieved at the node from which it was launched. "Same or Diff?" indicates there is no requirement on the launch and retrieval nodes for each drone trip. The last column of Table 3 shows the number of references that locate each type of facility, with aerial (UAV) and ground (UGV) drones indicated before and after the slash, respectively. Some references include more than one model for delivery variations or for locating various types of drones facilities, and thus contribute more than once in Table 3. The last row of the table shows the majority of research has addressed discrete space problems for UAVs with cost or time objectives and the same launch and retrieval points. This table shows that the greatest concentration of research has been on discrete models for locating DBs in systems with aerial drones. A second area of research concentration has been on problems where recharging stations are located. Table 3 also shows limited research on locating temporary drone bases (DB-T), customer points (CP), platooning points (PP), and (re)supply points (SP). Also, we found no study that investigates the location of the holding points (HP) for drone delivery problems.

We close this section with a categorization of the literature that provides a road map for the subsequent review. Fig. 5 shows our scheme that first divides the literature in terms of the type of vehicles used, namely research that uses only DVs, uses both DVs and OV's, or uses DVs (and also perhaps OV's) in novel applications. The boxes in Fig. 5 include the name (i.e., problem type), followed by one to three items in parentheses to indicate the number of references in that category, the relevant section of this paper, and the relevant summary tables in the paper. Row 3 of Fig. 5 subdivides the literature on problems where only DVs are used based on the facility type located (either only DB facilities or other facilities), and subdivides the literature on problems where both DVs and OV's are used based on the type of interaction between DVs and OV's (either no interactions, OV's resupplying DVs, or DVs (re)supplying OV's).

3. A general methodological framework

This section provides a general methodological framework for drone delivery facility location problems, with an illustration based on the model for the maximum coverage capacitated facility location problem with range constrained drones (MCFLPD) in Chauhan et al. (2019). Drone delivery problems share a number of common elements that reflect real-world operations and limitations, modeling assumptions, and logical requirements. These elements (e.g., drone range) can be modeled in a wide variety of ways, and there are many different operating assumptions incorporated in drone delivery models. Because

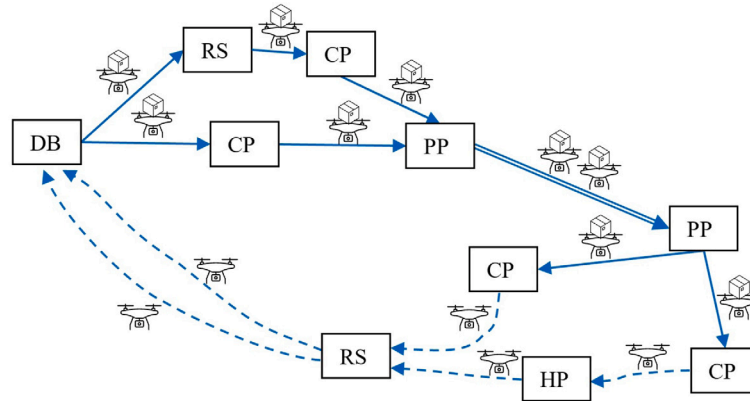


Fig. 2. Drone-only delivery network with DB, CP, RS, PP and HP.

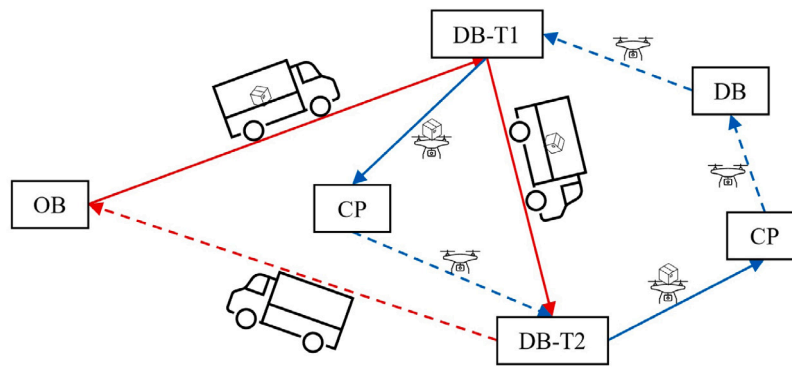


Fig. 3. Delivery network with truck and drone and DB-T.

Table 3
Number of references by facility type.

	Space		Objectives				Launch and Retrieval			UAV/UGV
	Discrete	Continuous	Cost	Time	Cover	Other	Diff.	Same	Same or Diff.	
Drone Base (DB)	48	7	31	19	10	9	4	48	3	48/7
Customer Point (CP)	3	0	3	0	0	1	2	1	0	3/0
Recharging Station (RS)	14	1	8	5	2	2	3	11	1	14/1
Holding Point (HP)	0	0	0	0	0	0	0	0	0	0/0
Platooning Point (PP)	6	1	7	0	0	0	4	3	0	0/7
Other Vehicle Base (OB)	2	1	3	0	0	1	0	3	0	2/1
Temporary Drone Base (DB-T)	4	0	1	2	0	1	1	1	2	3/1
(Re)supply Point (SP)	8	0	3	4	1	1	3	5	0	6/2
Sum	85	10	56	30	13	15	17	72	6	76/19

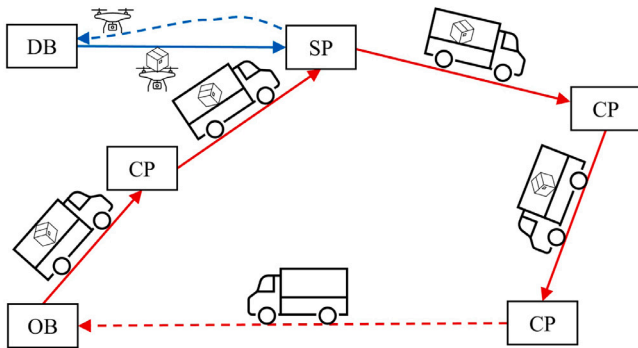


Fig. 4. Delivery network with truck and drone and SP.

models are generally not based on observation of existing drone delivery systems, researchers have had great latitude in choosing which features to include, how to model them, and which features to ignore.

To provide a general framework, we identify ten main elements common to a basic drone delivery model where drones (DVs) and/or trucks (OVs) deliver packages to a given set of customers. We discuss each element and illustrate how each element is handled in the MCFLPD model in Chauhan et al. (2019). In the MCFLPD model, range-limited DVs are launched from DBs to make deliveries to CPs by several one-to-one trips. For some of the elements in this section, we also include examples of how they are implemented differently in other studies. For the description of the sets, parameters and decision variables used in the formulation of the MCFLPD, see Table 4.

We present the description of the ten main elements (numbered i - x) below:

- (i) *Decisions on locations of facilities of various types (see Table 3 for facility types)*

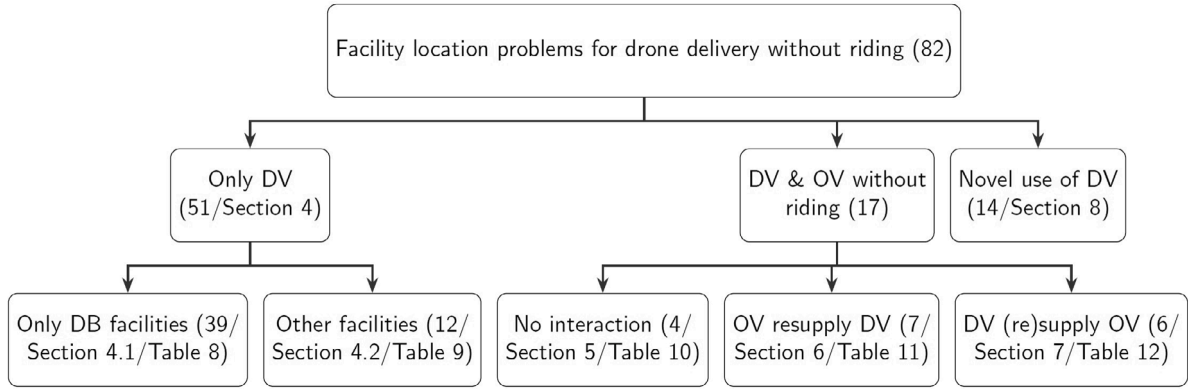


Fig. 5. The categorization of the literature.

Table 4

Notation used in the MCFLPD (Chauhan et al., 2019).

Notation	Definition
I	set of demand locations
J	set of potential facility locations
K	set of available drones
η	power transfer efficiency
θ	lift to drag ratio
B	battery capacity of each drone
b_{ij}	battery consumed during one trip between demand point $i \in I$ and facility $j \in J$
d_{ij}	travel distance between demand point $i \in I$ and facility $j \in J$
m_b	UAV battery mass
m_t	UAV mass tare, without battery and load
p	maximum number of facilities
U	capacity of each located facility
w_i	demand at location $i \in I$
x_{ijk}	a binary decision variable that equals one if customer $i \in I$ is served by drone $k \in K$ of facility $j \in J$
y_j	if the facility is located at $j \in J$
z_{jk}	if drone $k \in K$ is assigned to facility $j \in J$

In the MCFLPD formulation, the y_j variables indicate the facility location decisions for DBs.

(ii) *Routing decisions for the DVs and OVs (if any) from a starting depot to an ending location*

In the MCFLPD formulation, the x_{ijk} variable covers the DV “routing” decision on delivery to customer i from facility j by drone k . Because drone delivery models often involve vehicle routing for OVs, and sometimes DVs, in a similar way to non-drone routing problems, we do not cover the vehicle routing aspects in this review.

(iii) *Decisions on which DV or OV (if any) delivers to each customer*

In the MCFLPD formulation, the x_{ijk} variable captures the decision on which DV k delivers to a customer at i .

(iv) *One or more objectives (e.g., minimize cost, minimize completion time, maximize demand served, etc.)*

The MCFLPD aims to maximize demand coverage, so it sums the total demand delivered across all drones, customers and facilities:

$$\max \sum_{i \in I} \sum_{j \in J} \sum_{k \in K} w_i x_{ijk} \quad (1)$$

An important part of the drone delivery literature considers cost and time-based objective functions. For the former, a general cost-based objective function for drone delivery would minimize the sum of fixed costs for DB j (e_j), fixed costs for DV k (a_k) and the transportation cost of the DV traveling between DB j and

CP i . If $(f_{ij}(\cdot))$ represents the transportation cost, then such an objective is:

$$\min \sum_{j \in J} e_j y_j + \sum_{k \in K} a_k u_k + \sum_{i \in I} \sum_{j \in J} f_{ij}(\cdot) r_{ij} \quad (2)$$

where binary variables u_k and r_{ij} indicate whether DV k is used and whether CP i is assigned to DB j , respectively. Cost function $f_{ij}(\cdot)$ could be something simple such as $c_d \times d_{ij}$ (e.g., [Macias et al., 2020](#); [Xia et al., 2023](#); [Zhu et al., 2022](#)), similar to transportation costs in classical facility location problems. Some drone delivery research uses more complex nonlinear cost functions that depend on drone travel distance, payload and speed to model energy consumption of drones (e.g., [Dukkanci, Kara, & Bektaş, 2021](#); [Zhu et al., 2022](#)). [Dukkanci et al. \(2021\)](#) includes the following cost component related to drone energy consumption in the objective function:

$$\min \sum_{i \in I} \sum_{j \in J} \left[2\mu_1 \frac{d_{ij}}{v_{ij}} + 2\mu_2 d_{ij} v_{ij} + \frac{\mu_3^1 d_{ji} + \mu_3^2 d_{ij}}{(v_{ij})^2} + 2\mu_4 d_{ij} (v_{ij})^2 \right] \quad (3)$$

where parameters μ_1 , μ_2 , μ_3^1 , μ_3^2 and μ_4 are calculated based on the power consumption formula given by [Zeng, Xu, and Zhang \(2019\)](#). Here, μ_3^1 includes the package mass w_i for the trip of the loaded DV to the CP, and μ_3^2 does not include the package mass as it is the empty trip back to the DB. Similar expressions can be used for time-based objective functions. Some studies consider straightforward functions in minimizing the total travel time or minimizing the completion time of deliveries (makespan), while other researchers consider more complicated nonlinear functions. For instance, [Gentili, Mirchandani, Agnetis, and Ghelichi \(2022\)](#) uses a disutility function $g_i(\cdot)$ based on item perishability for CP i and is:

$$\min \sum_{j \in J} \sum_{i \in D(j)} \sum_{t \in T} \left(g_i \left(t - \frac{q_{ji}}{2} \right) s_{jti} \right) \quad (4)$$

where t is the delivery time slot, $\left(t - \frac{q_{ji}}{2} \right)$ represents the number of time slots required for the delivery to CP i from DB j , and s_{jti} is a binary variable indicating if the DV returns to DB j at the end of time slot t after serving CP i . $D(j)$ is the set of CPs for DB j , and T is the set of time slots. In computational experiments, the disutility $g_i(\cdot)$ is a quadratic function of the time.

(v) *Requirement that each customer receives a delivery, except in some models, e.g., where demand coverage is the objective*

This element is to ensure each customer receives a delivery; though this is not needed in all models, as for example when maximizing profit or demand coverage, like in the MCFLPD. A

variation of this element is to define a subset of customers that require delivery. With the MCFLDP, this could be

$$\sum_{j \in J} \sum_{k \in K} x_{ijk} = 1 \quad \forall i \in I' \quad (5)$$

- (vi) *Requirements that facilities of the appropriate type must be opened if used by a DV or an OV (if any)*

This element links the DV and/or OV vehicle routes and facility locations. Constraints (6) and (7) in the MCFLPD formulation ensure that each DV is assigned to at most one opened DB.

$$z_{jk} \leq y_j \quad \forall j \in J, k \in K \quad (6)$$

$$\sum_{j \in J} z_{jk} \leq 1 \quad \forall k \in K \quad (7)$$

- (vii) *Limitations on the number of open facilities (e.g., based on a given number or budget)*

This element limits the number of facilities that are open. The MCFLPD formulation addresses this with a p -median type constraint:

$$\sum_{j \in J} y_j \leq p \quad (8)$$

More generally, there could be a budget constraint with budget M , such as

$$\sum_{j \in J} e_j y_j + \sum_{k \in K} a_k u_k + \sum_{l \in L} e_l^R y_l^R \leq M \quad (9)$$

where, as before, e_j is the fixed cost for DB j and a_k is the fixed costs for DV k . Here, e_l^R is the fixed cost for RS $l \in L$ and y_l^R is a binary decision variable that equals one if an RS is located at $l \in L$ to extend the range of DVs. This element also addresses the issue mentioned in Section 2.1, of different facility types having different requirements and costs.

- (viii) *Limitation(s) on the drone travel range, which may be based on distance, time or energy (and can be nonlinear functions of drone speed)*

This element is an important feature for most types of DVs and models generally enforce a DV range limitation. Studies usually model the drone range in terms of time (endurance) (e.g., Bakach, Campbell, & Ehmke, 2021; Cokyasar, Dong, Jin, & Verbas, 2021; Pina-Pardo, Silva, & Smith, 2021) or distance (e.g., Gentili et al., 2022; Hong, Kubly, & Murray, 2018; Pinto & Lagorio, 2022; Shavarani, Mosallaeipour, Golabi, & İzbirak, 2019) based on an assumption or a model of energy (power) consumption and battery/fuel capacity. However, a few studies in the literature (Bruni & Khodaparasti, 2022; Bruni et al., 2023; Chauhan et al., 2019; Zhu et al., 2022) use ad-hoc energy consumption models from the literature to define the DV range. In particular, the MCFLPD uses a power consumption function proposed by Figliozzi (2017) to model the range of the DV, where power consumption b_{ij} between CP $i \in I$ and DB $j \in J$ is calculated by the following equation:

$$b_{ij} = \frac{m_i + m_b + w_i}{\theta \eta} d_{ji} + \frac{m_i + m_b}{\theta \eta} d_{ij} \quad \forall i \in I, j \in J \quad (10)$$

The first term of the right-hand side of the equation represents the forward journey from DB j to CP i of a loaded DV carrying a mass of w_i and the second term represents the return journey from CP i to DB j of an empty DV. In Eq. (10) the power transfer efficiency η and the lift-to-drag ratio θ represent complicated aspects that encompass the battery and energy system of the drone, the design of the drone including the rotors and/or propellers, the operation of the drone and the operating environment. The

MCFLPD formulation includes the following constraint to ensure the range of each DV is not exceeded.

$$\sum_{i \in I} b_{ij} x_{ijk} \leq B z_{jk} \quad \forall j \in J, k \in K \quad (11)$$

For a more detailed (and complicated) approach to modeling energy consumption during each phase of a delivery trip, see Appendix A.

- (ix) *Limitation(s) on the amount carried by the vehicles, which may be in units of packages, weight or volume*

This element is important for DVs, as these generally have small payload capacities of one or a few packages. The capacity may be expressed in abstract units (e.g., packages) or more explicitly in terms of the maximum volume and/or mass carried. The differences between DVs and OVs in terms of payload limitations are discussed in Section 2.1. Like most studies in the literature, Chauhan et al. (2019) assumes only one delivery per trip, and that the demand at any location is less than the capacity of the DV. Therefore, The MCFLPD formulation does not have a DV capacity constraint, but it does include

$$\sum_{j \in J} \sum_{k \in K} x_{ijk} \leq 1 \quad \forall i \in I \quad (12)$$

which ensures that each customer is visited by at most one drone. Some other studies explicitly include the capacity of the DV in their formulation with multiple deliveries allowed per drone trip (e.g., (Liu et al., 2020, 2021; Luo, Gu, Poon, Liu, & Lim, 2022; Uhm & Lee, 2022; Xia et al., 2023) For example, Xia et al. (2023) use C as the payload capacity of the DV and include the constraint:

$$\sum_{i \in I} \sum_{l \in L} w_i \chi_{ilk} \leq C \quad \forall k \in K \quad (13)$$

where χ_{ilk} is a binary variable indicating DV k travels from node i to node l .

- (x) *Other limitation(s) on DVs, OVs, and facilities such as DBs, RSs or CPs*

The final element covers other types of limitations on vehicles and facilities. Chauhan et al. (2019) includes the capacity of the DB as an additional constraint:

$$\sum_{i \in I} \sum_{k \in K} w_i x_{ijk} \leq U y_j \quad \forall j \in J \quad (14)$$

Other studies in the literature consider different types of limitations on DVs, OVs, and facilities. For instance, Cokyasar et al. (2021) includes a limitation on the RS facilities modeled as M/D/1 queues. The objective function of the problem includes a cost component related to the lead time and waiting time in a queue due to recharging (battery swapping) operations. In order to model queuing at RSs, the proposed mixed-integer nonlinear programming (MINLP) formulation includes the following two constraints to calculate the utilization rate and the average number of DVs waiting in a queue of an RS:

$$\rho_j = \psi^{-1} \sum_{l \in J_j \setminus \{V\}, i \in I} \lambda_i (y_{lji}^+ + y_{jli}^-) \quad \forall j \in J \quad (15)$$

$$L_j^q = \frac{\rho_j^2}{2(1 - \rho_j)} \quad \forall j \in J \quad (16)$$

where ρ_j is a continuous nonnegative variable that represents the utilization rate at RS location j , ψ is the service rate of an RS, λ_i is the demand rate at CP i , y_{lji}^+ and y_{jli}^- are binary variables indicating whether a DV flies from l to j to satisfy the demand at CP i and then back to l , respectively. Set $J_j \setminus \{V\}$ is the subset of neighboring vertices of a given vertex $j \in J$ that can be visited without needing a recharge, and L_j^q represents the average number of DVs at RS j waiting in the queue for

recharge (battery swap). Some studies include time windows (deadlines) for operations at facilities such as CPs (Alfandari, Ljubić, & da Silva, 2022; Bakach et al., 2021; Xia et al., 2023). For instance, Xia et al. (2023) includes the following constraint in their formulation to ensure that the time window of each CP will not be violated:

$$t_i^{\min} \leq t_{ik} \leq t_i^{\max} \quad \forall i \in I, k \in K \quad (17)$$

where t_{ik} is a continuous nonnegative variable to represent the arrival time of DV k at CP i , and $t_i^{\min}$ and $t_i^{\max}$ are lower and upper bounds of time window for CP i . In addition to the limitations discussed above, a few studies consider obstacle-avoiding paths for DVs (Hong, Kuby, & Murray, 2017; Hong et al., 2018).

We remark that many of the ten elements above are common to classic facility location models (p-median, set covering, etc.) or location-routing models; and some drone delivery research has directly used well-known facility location models, just with drones providing transportation (e.g., Kim, Lim, Cho, & Côté, 2017 use a simple set covering model). However, as discussed in Section 2.1 there are important features for facility location drone delivery problems that differ from classic facility location problems. Further, in drone delivery problems where DVs ride on OV (see Dukkanci et al., 2023), there are additional considerations for the synchronization of OVs and DVs, similar to those for synchronized VRPs (Drexler, 2012).

The following five sections review the literature on facility location for drone delivery, following the categorization presented in Fig. 5. In these sections we use our consistent set of facility and vehicle types (e.g., DB, OB, DV, etc.), but when appropriate we also indicate in parentheses the particular name used in the reference (e.g., “DB (drone depot)” or “OV (truck)”).

4. Problems with only drones

The section reviews research on problems where drones (DV) are the only vehicles. In these problems, the facilities to be located are DBs, RSs, and DB-Ts. Section 4.1 discusses research that locates only DB facilities, while Section 4.2 discusses research that locates RS and DB-Ts, sometimes along with DBs.

4.1. Problems with location of only DB facilities

This section reviews delivery applications where the network consists of only drone bases (DB) and customers (CP), and the locations of one or more DBs are to be decided. Some examples of DB names from the literature are drone centers, distribution centers, facilities, hospital hubs, drone stations, depots, drone-beehives, drone logistics hubs, and delivery centers. This subsection groups the problems in terms of their objective function, and we review the studies in the order: (i) cost-based objectives, (ii) time (distance)-based objectives and (iii) other types of objective functions (e.g., coverage-based and vehicle resource-based objective functions).

Tables 8 presents a summary of the 39 studies with only drone vehicles (DV) which locate only DB facilities. The table is organized as follows. The first column provides the reference. The second column addresses the model objective, where the meaning of the letters is explained in Table 5. Column three indicates the modeling space, either discrete (D) or continuous (C). Column four shows the drone type as either aerial (A) or ground (G). Column five indicates the number of customers visited per drone route, and the following column indicates whether or not customer time windows (TW) and uncertainty (U) are included. Due dates or deadlines for CPs are also considered as time windows. Column seven indicates the application area, either commercial (Com), humanitarian (Hum) or healthcare (Hea). Column eight indicates the formulation type given in Table 6, and the last

Table 5

Acronyms related to objective function.

Objective	Argument	
C(.,.) = Cost-based	a	fixed cost of a facility
	b	fixed cost of DVs
	c	variable cost of DVs
	d	variable cost of OVs
	e	other cost components
T(.,.) = Time-based	f	total travel time
	g	completion time (makespan)
	h	other time related components
V = Coverage-based	–	
O = Other	–	

Table 6

Acronyms related to formulations and features.

Acronym	Meaning
IP	Integer Program
MILP	Mixed-Integer Linear Program
MINLP	Mixed-Integer Nonlinear Program
MISOC	Mixed-Integer Second-Order Cone Program

Table 7

Acronyms related to solution approach.

Acronym	Meaning
CS	Commercial solver
BP	Branch-and-price algorithm
BD	Benders decomposition
CP	Cutting plane
BBC	Branch-and-Benders cut algorithm
BC	Branch-and-cut algorithm
DDD	Dynamic discretization discovery
CCG	Column-and-constraint generation algorithm
PSH	Problem specific heuristic
SO	Sequential optimization
LNS	Large neighborhood search
BDH	Benders decomposition-based heuristic
BPH	Branch-and-price algorithm-based heuristic
GA	Genetic algorithm
MA	Memetic algorithm
SA	Simulated annealing
MatH	Matheuristic
AA	Approximation algorithm
GH	Greedy heuristics
NSGA	Non-dominated sorting genetic algorithm
CSA	Cuckoo search algorithm
GRASP	Greedy randomized adaptive search procedure
CA	Continuous approximation
ILS	Iterated local search
AIA	Artificial immune algorithm
TS	Tabu search
ACO	Ant colony optimization

column shows the solution approaches as exact (E) and/or heuristic (H) with the acronyms explained in Table 7.

Table 8 indicates that most of the 39 articles consider discrete space (34) where aerial drones (38) operate to serve a single customer per trip (29). Time window restrictions (3) and uncertainty issues (10) are rarely addressed. On the other hand, the literature considers different types of objective functions, namely cost (19), time (14) and coverage-based (10) functions, as well as a range of application areas, including commercial (14), healthcare (16) and humanitarian (9) applications.

Studies with UAVs in discrete space

Most of the studies in this category consider a cost-based objective function. Some studies present a sequential optimization approach where the first model minimizes either the fixed cost of DBs or battery consumption, and the second model minimizes the operating cost of DVs, number of DVs, or a cost function. Kim et al. (2017) is one of

Table 8

Classification of literature on DB location problems with only drones.

	Objective	Space	DV	# customers	Feature	App	Form.	Sol App E/H
Kim et al. (2017)	C(a,b)	D	A	>1		Hea	IP	-/SO
Torabbeigi et al. (2020)	C(a),O	D	A	>1		Com	MILP	-/SO
Macias et al. (2020)	C(a,c,e)	D	A	>1		Hum	MILP	CS/LNS
Kim, Lee, and Moon (2019)	C(a,e)	D	A	1	U	Hum	IP	-/BDH
Otero Arenzana, Escibano Macias, and Angeloudis (2020)	C(a,b,c,e)	D	A	1		Hea	MILP	CS/-
Zhu et al. (2022)	C(a,c,e)	D	A	1	U	Hum	MILP	CCG,BD/CCG
Xia et al. (2023)	C(c,e)	D	A	>1	TW	Com	MILP	BP/-
Cokyasar, Stinson, Sahin, Prabhakar, and Karbowski (2023)	C(a,c)	D	A	1		Com	MILP	-/MatH
Park, Lee, Sung, Nielsen, and Moon (2022)	C(a,b)	D	A	1	U	Hea	MILP	BP/BPH
Bruni et al. (2023)	C(b,c,e)	D	A	>1		Com	MILP	CS/-
Golabi, Shavarani, and Izbirak (2017)	T(f)	D	A	>1	U	Hum	MINLP	-/GA,MA,SA
Uhm and Lee (2022)	T(f)	D	A	>1		Com	MILP	CS/PSH,TS
Bruni and Khodaparasti (2022)	T(f)	D	A	>1	U	Com	MINLP	CS/MatH
Gentili et al. (2022)	T(h)	D	A	1		Hum	MILP	CS/-
Ghelichi, Gentili, and Mirchandani (2022)	T(h),O	D	A	1	U	Hum	MINLP	-/AA
Wang, Jiang, and Qi (2022)	T(h)	D	A	1		Hum	MISOC	CS/-
Chauhan et al. (2019)	V	D	A	1		Hum	MILP	CS/GH
Chauhan, Unnikrishnan, Figliozzi, and Boyles (2021)	V,O	D	A	1	U	Hum	MILP	CS/GH
Al-Rabiaah, Hosny, and AlMuhaideb (2022)	V	D	A	1		Hea		-/GH
Pinto et al. (2020)	V	D	A	1*		Com	MILP	CS/-
Shavarani, Golabi, and Izbirak (2021)	C(a,b),V	D	A	1		Com	MINLP	-/NSGA II&III
Rashidzadeh, Hadji Molana, Soltani, and Hafezalkotob (2021)	C(a,b,c),V,O	D	A	1		Hea	MILP	CS/-
Shi, Lin, Li, and Li (2022)	C(a,b,c),T(f)	D	A	>1	TW	Hea	MILP	-/NSGA II
Wang, Zhao, Cheng, and Qi (2023)	C(a,b),T(f,h)	D	A	1	U	Hea	MISOC	CS/-
Claesson et al. (2016)	T(f)	D	A	1		Hea		-/AA
Pulver, Wei, and Mann (2016)	V	D	A	1		Hea	IP	CS/-
Boutillier et al. (2017)	O	D	A	1		Hea	IP	CS/-
Pulver and Wei (2018)	V	D	A	1		Hea	MILP	CS/-
Mackle et al. (2020)	T(f)	D	A	1		Hea		-/GA
Wankmüller et al. (2020)	T(f),O	D	A	1		Hea	MILP	CS/-
Feng, Murray, and Church (2021)	T(f),V	D	A	1		Hea	IP	CS/-
Boutillier and Chan (2022)	T(f),O	D	A	1	U	Hea	MILP	CS/-
Chauhan, Unnikrishnan, Figliozzi, and Boyles (2023)	V	D	A	1	U	Hea	MILP	CS/-
Aurambout, Gkoumas, and Ciuffo (2019)	C(a,b,c,e)	C	A	1		Com		-/SO
Aurambout, Gkoumas, and Ciuffo (2022)	C(a,b,c,e)	C	A	1		Com		-/SO
Pan, Song, Chu, and Peng (2020)	C(e)	C	A	1		Com		-/CSA
Perera, Dawande, Janakiraman, and Mookerjee (2020)	C(a,c,e)	C	A	1		Com		CS/-
Liu (2023)	T(h)	C	A	1		Com	MINLP	-/PSH
Sonneberg, Leyerer, Kleinschmidt, Knigge, and Breiter (2019)	C(b,c,e)	D	G	>1	TW	Com	MILP	CS/-

C(.,.) is a cost objective, T(.,.) is a time objective, V is a coverage objective and O is other objectives, where details are provided in Table 5.
1*: there is also a pickup point where the drone flies to in order to load the package.

the first studies that considers locating DBs (drone center locations) selected from a discrete set of points. The authors propose a delivery and pickup system where DVs are used to deliver medications to CPs (patients) and then pick up test kits to bring back to the DB. The solution approach includes: (i) a set-covering problem to find the number and locations of DBs while minimizing the fixed setup cost of DBs, and (ii) a multi-depot vehicle routing problem with pickup and delivery to decide the number of DVs and their routes, while minimizing the operations cost of DVs. The authors propose preprocessing rules and two bound generation methods, including a partition method to find upper bounds and Lagrangian relaxation to obtain lower bounds. Torabbeigi et al. (2020) extends the study of Kim et al. (2017) by considering the battery consumption rate as a function of payload. The authors do not propose the problem as a healthcare application, but they do propose solution via (i) a set-covering problem to find locations of DBs, and (ii) a vehicle routing problem to determine the number of DVs and their routes while minimizing the number of DVs. Macias et al. (2020) presents a different two-stage approach consisting of trajectory optimization and a depot location and routing problem. The former is an optimal control problem, which is transformed into a nonlinear program, defined to minimize battery consumption while considering aerodynamics parameters, battery properties and topography. The latter problem decides the location of DBs (hubs), DV routes, and battery management while minimizing the sum of fixed costs of DB setup, travel time cost, penalty cost for unsatisfied or oversatisfied demand, and fixed battery cost. The authors develop a large neighborhood search solution algorithm.

The remaining studies in this section that present discrete models with a cost-based objective minimize a “total cost” expression that includes the operating cost of DVs, along with one or more other costs.

Kim et al. (2019) consider uncertainty while locating DBs in a humanitarian application to minimize the sum of fixed DB costs and operation and maintenance costs of DVs. Uncertainty of the flight distance is considered by defining a chance constraint to model possible fluctuations of energy consumption during the DV flight. The problem is defined on a multi-period planning horizon. A Benders decomposition-based heuristic solution algorithm is developed. Otero Arenzana et al. (2020) design a hospital delivery network where some hospitals are selected as DBs (hospital hubs) to store and deliver blood to CPs (other hospitals) using DVs. The objective function minimizes the sum of: (i) generalized costs per time covering typical logistics costs, (ii) battery and electricity costs covering DV energy costs for traveling, (iii) fixed costs of purchasing DVs, (iv) fixed costs setting up DBs, and (v) carbon emission costs, including for battery production and electricity generation. The model also restricts the number of DBs to be operated. The authors develop a mixed-integer linear programming (MILP) formulation and present a case study based on National Health Service hospitals in London. Zhu et al. (2022) study the two-stage robust facility location problem with DVs for humanitarian relief after a disaster under demand uncertainty. In the first stage, the location of DBs (facilities) and the assignment of DVs to DBs is determined. In the second stage, the assignments of CPs (customers) to DBs, and DVs to CPs are determined. The objective function minimizes the sum of fixed cost of opening DBs, transportation cost, and penalty costs for the

unsatisfied demand. The authors propose three different models, where the first two explicitly calculate drone energy consumption depending on the payload estimated by a non-linear model (Liu, Sengupta, & Kurzhanskiy, 2017) and a linear model (Figliozzi, 2017), respectively. The third model allows split deliveries, and uses a simplified version of the energy consumption model in Figliozzi (2017). The authors develop column-and-constraint generation and Benders decomposition methods for the third model, and a heuristic version of the column-and-constraint generation method for the first two models. Xia et al. (2023) study a drone routing problem with a blockchain-enabled fleet, where drone operators collaborate to share all the information about DV fleets and their conditions using blockchain technology. This collaboration helps to reduce the negative impacts of battery wear and disposal, as DVs are allowed to return to DBs of other operators for reloading and recharging. The objective is to minimize the sum of battery-related costs and the transportation cost of DVs while satisfying time window and capacity constraints. A branch-and-price algorithm is developed. Cokyasar et al. (2023) analyze regional impacts of direct drone delivery in terms of energy consumption and compare it to the impact from deliveries made by both diesel and battery electric trucks. The authors propose a capacitated facility location problem to locate DBs (fulfillment centers) while minimizing the total cost including the fixed cost of DBs and energy costs of DVs. To calculate the energy consumption of DV, the authors utilize the findings of Kirschstein (2020) and they also model different types of drones and the impact of wind speed. The authors proposed a MILP formulation to solve the problem. To solve a case study with hundreds of thousands of nodes based on Chicago, US, they apply several problem-specific size reduction techniques. Park et al. (2022) study a facility location-allocation problem for an emergency medical service where the coverage distance and capacity of DBs depend on the number of DVs assigned to DBs. The problem aims to minimize the total cost, including fixed costs of DBs and DVs, while accounting for the uncertainty of demand using a robust optimization framework. The authors first develop a MINLP formulation. They also employ a branch-and-price algorithm where the MILP formulation and decomposed dynamic programming are used to solve the subproblem. Additionally, the authors propose a restricted master heuristic to solve large-size instances. Bruni et al. (2023) propose the drone routing problem with shared depots as a location-routing problem where the location of DBs and fleet size and routing of DVs are determined while minimizing the sum of the transportation cost, fleet usage cost and weight-based cost related to packaging and handling at DBs. In this problem, a non-linear load-dependent energy consumption model proposed by Dorling, Heinrichs, Messier, and Magierowski (2017) is considered. The authors develop a MILP formulation without approximating non-linear terms of the energy consumption model.

Six studies in Table 8 consider locating DBs to minimize the delivery time. Golabi et al. (2017) study a stochastic facility location problem to determine the location of a fixed number of DBs (relief distribution centers). It is assumed that affected people are uniformly distributed along the network edges and the demand of each point on the edge is covered by the closest open DB. If the nodes of an edge have different closest open DBs, then the edge is partitioned into two segments, each of which is assigned to its closest open DB. Due to the impact of the disaster, some edges of the network can be inaccessible. If an edge is accessible, then it is assumed that the affected population will travel to DBs, and if an edge is inaccessible, then DVs will be used to drop relief kits at equal distances along the inaccessible edge. The objective function is to minimize the aggregate traveling time of both the people who live on accessible edges and the DVs (to cover people on inaccessible edges). The problem also includes uncertainty of demand and accessibility of the network. The authors propose a genetic algorithm to solve this problem and conduct a case study in Tehran, Iran based on earthquake scenarios. Uhm and Lee (2022) is the only study that considers a safe separation distance

between DVs in a multi-depot VRP where multiple DVs are operating. The problem minimizes the total operation time of DVs including their return journey to their DBs while also satisfying capacity and maximum travel time constraints. The authors first consider a time-based model, where the location of each DV is calculated at discrete time intervals, resulting in a MINLP. Then the authors propose a scheduling-based model, where departure times of DVs are scheduled to ensure that the distance between DVs is greater than the separation distance. The authors present a MILP for the second model. The authors develop two solution approaches: (i) a heuristic that decomposes the problem into routing and scheduling subproblems; and (ii) a hybrid tabu search heuristic. Bruni and Khodaparasti (2022) consider a shared delivery system, similar to Xia et al. (2023), where DBs (drone beehives) are shared by different freight carriers. They define the problem as a location-routing problem to decide the locations of DBs and the routes of DVs. DVs are allowed to return to any DB after completing their routes. The nonlinear energy consumption model in Dorling et al. (2017) is used to define the drone range and travel time uncertainty is addressed via a robust framework while aiming to minimize the total waiting time of customers. The authors present an MINLP, and a matheuristic combining a variable neighborhood descent heuristic with a cutting plane approach. Gentili et al. (2022) study the drone location and scheduling problem where perishable medical products are delivered by DVs to hard-to-access areas during a humanitarian disaster. DVs operate from DBs (platforms) and have a limited range. The first variant of the problem decides the locations of DBs, the assignments of CPs (demand points) to DBs and the order of visiting CPs while minimizing a total disutility function that measures customer disutility based on time slots. The second problem variant proposes a two-period dynamic location problem, where DBs can be relocated during the night and could be considered as DB-Ts. The authors propose a MILP based on the time-slot version of both problem variants, where time is discretized into slots. Ghelichi et al. (2022) extends the study of Gentili et al. (2022) by considering the uncertainty of demand points while locating DBs (platforms). The objective function minimizes a total disutility function consisting of total delivery time and a penalty for uncovered demand points. The authors propose a chance-constrained programming formulation and an approximation method. Wang et al. (2022) study a facility location problem for drone-based emergency deliveries where the system congestion of DV requests is considered via a queuing model with three different queuing priority disciplines. For each discipline, the authors approximate the models as mixed-integer second-order conic programs to minimize the longest expected response time, which includes the expected waiting time and the travel time.

Several studies aim to maximize the demand coverage. Chauhan et al. (2019) propose the maximum coverage facility location problem with DVs to locate a fixed number of DBs (facilities), from which range-constrained DVs are launched. The recharging of DV batteries is ignored. The authors present a novel MILP model to allocate DVs to DBs, and allocate CPs to DVs, while satisfying range restrictions on DVs based on an energy consumption model from Figliozzi (2017). Greedy and three-stage heuristic approaches are used to solve the problem. Chauhan et al. (2021) presents an extension to the problem by considering uncertainties of initial battery availability and battery consumption using a robust approach. The objective function of the problem is updated by adding a penalty function associated with initial battery availability. Al-Rabiaah et al. (2022) develop a “maximum coverage greedy randomized heuristic” for the problem proposed by Chauhan et al. (2019). Pinto et al. (2020) study the online meal ordering and delivery-network design problem, where DVs take off from DBs (charging stations) and visit restaurants to pick up meals to deliver to CPs. Their model decides the location of DBs while maximizing covered demand, where a CP is covered if at least a specified number of restaurants can serve that CP within a service time-bound. The number of DBs is limited by a budget constraint. The authors present a MILP to solve the problem.

Some studies include more than one objective function to address the concerns of different stakeholders. [Shavarani et al. \(2021\)](#) presents a capacitated bi-objective location problem similar to that of [Golabi et al. \(2017\)](#) that considers uniformly distributed demand and edge partition assumptions. The two objective functions are to: (i) minimize the total fixed costs for DBs (facilities) and DVs, and (ii) minimize the total uncovered demand. The capacity of a DB is defined by the total traveled distances of the DVs assigned to that DB. To solve this bi-objective problem, the authors implement the non-dominated sorting genetic algorithm II (NSGA-II) and the reference-point based nondominated sorting genetic algorithm (NSGA-III). A case study is conducted in San Francisco, CA. [Rashidzadeh et al. \(2021\)](#) extend the problem in [Chauhan et al. \(2019\)](#) where the authors study a multi-objective facility location problem for a blood supply chain by considering economic, environmental, and social impacts of last-mile deliveries by DVs. One objective is to maximize demand coverage to account for the social impacts of deliveries. The other two objectives are to: (i) minimize the total cost, including the fixed costs for DBs (blood distribution centers), fixed costs for DVs, and transportation cost of DVs based on energy consumption, and (ii) minimize the total CO₂ emissions at electricity generation facilities to power the DVs. The authors develop a preemptive fuzzy goal programming approach to solve the model. [Shi et al. \(2022\)](#) propose a bi-objective multi-trip drone location routing problem with simultaneous pickup and delivery for a healthcare application where the locations of DBs (central hospitals) and the routes of DVs are decided, with constraints on the payload and energy capacity of the DV, and delivery times to the CPs (epidemic prevention nodes). The two objective functions are to: (i) minimize the total cost including the fixed cost of opening DBs and DVs, and the travel and energy cost of DVs, and (ii) minimize the total working hours of all DVs. The authors proposed a modified NSGA-II to solve the problem. [Wang, Zhao, Cheng, and Qi \(2023\)](#) study a multi-objective facility location problem with congestion and priority for drone-based emergency deliveries. The problem also considers uncertainties of DV endurance and the demand arrival rate under a priority (M/G/1) queuing strategy by a fuzzy approach. The problem determines the DB location, DV fleet size and demand assignment while considering three objective functions: (i) total cost consisting of the fixed cost of DBs and deployment cost of DVs, (ii) system efficiency defined by total (average) service time and (iii) equitable response defined by the maximum response time. The authors propose a crisp mixed-integer second-order conic programming by applying chance-constrained, second-order conic, fuzzy and weighted goal programming approaches. A case study based on Orange County, CA is conducted.

Studies for delivery of automated external defibrillators

An important body of literature on drone delivery problems with location decisions is based on the transportation of automated external defibrillators (AEDs) by DVs to people who suffer out-of-hospital cardiac arrest (OHCA). [Claesson et al. \(2016\)](#) is one of the first studies that consider the location of DBs, from which the DV delivers an AED to the location of an OHCA. The model aims to minimize the arrival time of DVs weighted by the density of previous OHCA cases. The optimal locations of DBs are found by Multi Criteria Evaluation using GIS. The author conducts a case study in Stockholm County, Sweden. This study is based on the bachelor thesis by [Lennartsson \(2015\)](#) and the PhD thesis by [Fredman \(2018\)](#). [Pulver et al. \(2016\)](#) consider the same problem as a covering problem to maximize the total demand coverage with a service time limit. The problem is formulated as an IP and solved by a commercial solver. The authors conduct a case study in Salt Lake County, Utah. [Boutilier et al. \(2017\)](#) extend the study of [Pulver et al. \(2016\)](#) by considering more details of DV operations, including the busy time of DVs. Previous studies assume that while a DV is busy, there will be no other OHCA cases. This study relaxes that assumption to have multiple DVs in each DB (base). The authors solve the problem in two stages: (i) an IP model minimizes

the number of DBs while deciding locations that respect coverage constraints, and (ii) a queuing model based on a continuous-time Markov Chain determines the number of DVs per DB. [Pulver and Wei \(2018\)](#) present an extension with backup coverage, where a backup DV is available if the primary DV is unavailable. Here the authors also account for the partial coverage of continuously distributed demand. The proposed problem has two objective functions: to (i) maximize the primary coverage, and (ii) maximize the backup coverage. [Mackle et al. \(2020\)](#) present another aerial ambulance drone problem, which aims to minimize response times for OHCA incidents while determining the number and location of DBs. The authors compare the proposed system with DVs to a traditional ambulance service and layperson rescuers. [Wankmüller et al. \(2020\)](#) study the optimal allocation of DVs with AED in mountainous regions. The authors consider infrastructural and geographical obstacles while calculating travel time between a DB and a patient. The problem minimizes the weighted sum of the number of DVs used and the average travel time of DVs while deciding allocations of DVs to DBs and the assignments of patients to DBs. [Feng et al. \(2021\)](#) consider spatiotemporal heterogeneity while locating DBs, so that the demand and travel time are varied over a continuous space and time. The objective is to maximize the demand service quality (measured by service time). The authors combine a raster-based travel cost approximation with an IP model. A case study is for delivery of AEDs in southern California. The authors also present an extension that considers the location decision over time. [Boutilier and Chan \(2022\)](#) extend the study of [Boutilier et al. \(2017\)](#) with an integrated location-queuing problem that combines the p-median problem and an explicit $M/M/d/d$ queuing model to minimize the total number of DVs required while ensuring a response-time goal and almost-full availability of a DV in the event of an OHCA. The authors reformulate the model using existing response times, which can then be solved by an optimization solver. They also employ a tactical simulation model to evaluate the performance of the network and a machine learning model to estimate survival rates based on response-time reductions. [Chauhan et al. \(2023\)](#) extends the problem in [Chauhan et al. \(2019\)](#) by including multiple periods and a coverage reliability constraint via a chance constraint for AED delivery. The authors develop a MILP formulation and conduct a case study in Portland, OR.

Studies with UAVs in continuous space

Five papers have studied drone delivery using continuous space modeling. [Aurambout et al. \(2019\)](#) study the market potential and the economic viability of DVs for last-mile deliveries in Europe. The authors present an approach using population to determine the location of DBs (multi-level fulfillment centers called drone-beehives). The proposed model estimates the number of people who could benefit from DV deliveries, as the main objective is to maximize the financial return. The model consists of two phases. The first phase estimates the maximum return of each possible location of DBs, and the second phase allocates DBs to potential CPs starting from the DBs with the highest potential return. To estimate potential return and economic viability, the authors consider DB cost, DV cost, and fixed and variable delivery costs. The results suggest 7% of the European population could receive such a service. [Aurambout et al. \(2022\)](#) extends this study with a different way to calculate the financial return, further analyses and applications to Europe and the USA. The results indicate that 16%–43% of the European population and 32%–60% of the USA population could use this service. [Pan et al. \(2020\)](#) presents a problem of locating DBs (drone logistics hubs) in remote rural areas to serve surrounding CPs (villages). The objective is to minimize a cost function that accounts for the distance between the DB and the CPs, the population of the CPs, and the difficulty of ground transportation (in comparison to air transportation by DVs). The problem is solved by a compact cuckoo search algorithm. [Perera et al. \(2020\)](#) analyze the impact of last-mile deliveries by DVs on the number of DBs (delivery centers) and delivery lead times using an idealized setting where CPs are uniformly located

Table 9

Classification of literature on other facility location problems with only drones.

	Facility	Objective	Space	DV	# customers	Feature	App	Form.	Sol App E/H
Hong et al. (2017)	RS	T(f),V	D	A	1		Com	MINLP	-/SA
Hong et al. (2018)	RS	V	D	A	1		Com	MILP	CS/SA
Pinto and Lagorio (2022)	RS	T(f),O	D	A	1		Com	IP	CS/Math
Ghelichi et al. (2021)	RS	T(f)	D	A	1		Hea	IP	CS/-
Huang and Savkin (2020)	RS	O	C	A	1		Com		-/PSH
Gómez-Lagos, Rojas-Espinoza, and Candia-Véjar (2022)	DB-T	T(g)	D	A	1*	TW	Hum	MILP	CS/GRASP
Shavarani, Nejad, Rismanchian, and Izbirak (2018)	DB, RS	C(a,b,c)	D	A	1		Com	MINLP	-/GA
Shavarani (2019)	DB, RS	T(f)	D	A	1		Hum	MILP	-/GA
Shavarani et al. (2019)	DB, RS	C(a,b,c)	D	A	1	U	Com	MINLP	-/GA
Dhote and Limbourg (2020)	DB, RS	C(a,c)	D	A	1*		Hea	IP	CS/-
Chalupa and Nielsen (2018)	DB, RS	C(a,c)	D	G	1		Com	MILP	CS/-
Cicek, Koç, Gultekin, and Erdoğan (2022)	DB-T, RS	T(f)	D	A	1	TW	Com	MILP	CS/GA

C(.,.) is a cost objective, T(.,.) is a time objective, V is a coverage objective and O is other objectives, where details are provided in Table 5.

1*: there is also a pickup point where the drone flies to in order to load the package.

on the circumference of the circle. The objective is to maximize the total profit while determining the number of DBs and delivery speed. To calculate profit, the problem includes the operating cost of DBs, the delivery cost of DVs, and ordering and holding costs of inventory. The authors derive analytical results showing decentralization of delivery networks as DVs become less costly, the benefits of limited delivery time customization, and benefits of faster delivery. Liu (2023) study an elliptical cover problem for drone delivery to locate DBs (depots) to minimize the maximum distance of a drone trip, which is defined as being from the nearest DB to a CP (demand point), then visiting the CP, and ending at the second nearest DB. The authors propose Cooper's alternating locate-allocate heuristic for the point-coverage and area-coverage scenarios.

Studies with UGVs

There is only one study in this subsection that focuses on ground drones (UGVs) and the problem is defined on a discrete space. Sonenberg et al. (2019) present a location routing problem with time windows where DVs (delivery robots) are routed between CP locations. The problem determines the locations of DBs (stations), CP assignments, the number of DVs, and routes of DVs while minimizing the total cost consisting of the transportation costs, rental costs of DVs, and personnel costs for loading operations. The problem also accounts for time window, payload and battery capacity constraints. The authors develop a MILP, which is solved by a commercial solver.

4.2. Problems with only drones and location of facilities other than just DBs

This subsection reviews the literature with drones as the only vehicles where facilities other than only DBs are located. We first cover research that locates RS facilities alone, then one study that locates DB-Ts, then research that locates RS along with other facilities. Table 9 shows the characteristics of 12 studies. This table uses the same format as Table 8, with the exception that a new column is added (the second column) to show the type(s) of facilities being located.

Table 9 shows that most of the 12 problems in this section consider a discrete space (11) where aerial drones (11) serve single customers per trip (12) in a commercial application (8). While the literature covers cost (4), time (6) and coverage (2) objectives, time window restrictions (2) and uncertainty issues (1) are rarely discussed.

Studies with UAVs where only RS are located

This subsection first analyzes the literature on problems with UAVs, in which the location of only RSs is determined in discrete space. Hong et al. (2017) is perhaps the first study that determines the locations of RSs (recharging stations) to extend the range of DVs to make deliveries to CPs. This is also the first study in the drone delivery literature that considers DV travel obstacles, and so an obstacle-avoiding path derivation technique is utilized. The authors develop a "distance-restricted coverage location model" that maximizes demand coverage

while minimizing the average flight distance from DBs (warehouses) to RSs. The authors propose a simulated annealing algorithm with a greedy approach. Hong et al. (2018) studies a variant of this problem, where the objective function maximizes the demand coverage. The problem considers two different coverage measures: (i) station-to-station coverage, and (ii) final demand coverage, which ensures safe return of a DV after a delivery. The authors develop a MILP formulation and a heuristic algorithm, similar to that in Hong et al. (2017). Pinto and Lagorio (2022) study a problem similar to the one presented by Hong et al. (2018). The authors optimize the number and location of RSs while minimizing two objective functions: (i) the total distance and (ii) the total number of RSs. Unlike (Hong et al., 2018), this problem does not consider obstacle-avoiding paths. Ghelichi et al. (2021) present the multi-stop drone location and scheduling problem for a healthcare and humanitarian application, where DVs are launched from DBs (urban providers of medical items), visit one or more RSs (charging platform) if necessary, and deliver medical items to CPs (rural or suburban clinics). The authors present a time-slot formulation that locates RSs, assigns CPs to DBs and schedules DV trips while minimizing the total completion time of all trips. The authors assume that RSs can be installed on existing structures such as streetlights, traffic lights and billboards and DVs can return to DBs from CPs without visiting any RSs, as the batteries of DVs are replaced at the CPs. The range of a DV is limited by its energy consumption and all CPs must be served within a daily operational period. The authors propose a preprocessing approach to eliminate dominated trips and solve the IP model. They conduct a case study in Louisville, Kentucky.

There has been some research on locating RS using continuous space models. Huang and Savkin (2020) extend the problem proposed by Hong et al. (2017, 2018), by considering a continuous space for the deployment of RSs (charging stations). The objective minimizes the number of RSs to be deployed given a demand coverage threshold. This study does not consider obstacle-avoiding paths for DVs. The authors develop a two-phase deployment algorithm, where RSs are first deployed in a triangle pattern, and then the RSs covering no demand, or the least demand, are removed and the remaining RSs are repositioned.

Studies with drones where only DB-T are located

Gómez-Lagos et al. (2022) study a pickup-delivery routing problem that locates only DB-Ts, where DVs operate in discrete space. The goal is to find a schedule of the DV path from a DB (depot) to a DB-T (facility) to pick up the package, which it then delivers to the CP. Then, the DV goes to the same or another DB-T to get the package for the next CP. When all deliveries are completed, the DV returns to the DB. The authors present three different MILP formulations to minimize the makespan of the drone fleet and develop a greedy randomized adaptive search solution procedure (GRASP).

Studies with drones where RS facilities and another facility type are located

The subsection first reviews the literature on problems with UAVs where both the location of DBs and RSs are determined. [Shavarani et al. \(2018\)](#) study a hierarchical facility location problem to decide the number and location of DBs (launch stations) and RSs (recharging stations) while minimizing the total cost including establishment costs of DBs and RSs, and purchasing and transportation costs of DVs. Similar to the study of [Golabi et al. \(2017\)](#), demand is uniformly distributed along the edges and satisfied by the closest open DBs. DVs are allowed to visit more than one RS to satisfy the demand. The problem is formulated as a MINLP and solved by a hybrid genetic algorithm. The authors conduct a case study based on Amazon Prime Air in San Francisco. [Shavarani \(2019\)](#) extends this study to a humanitarian application for relief distribution operations. The objective is to minimize the total traveled distance. A MILP formulation is presented and the problem is solved by a hybrid genetic algorithm. A case study in Tehran, Iran is conducted. [Shavarani et al. \(2019\)](#) present another extension of [Shavarani et al. \(2018\)](#) where the waiting of the customer is calculated based on a queuing system. Also, the authors consider uncertainty by introducing fuzzy variables for the demand, the costs of opening DBs and RSs, the purchasing cost of the DV and the distance limit (endurance) of the DV. Different than the previous two studies, demand is defined on a node, rather than along an edge. The objective is to minimize the total cost including the costs of opening DBs and RSs, and the purchasing and operating costs of DVs. [Dhote and Limbourg \(2020\)](#) design DV networks to deliver biological material between hospitals, laboratories and blood transfusion centers. The DV takes off from one of the DBs, flies to a pickup point to collect the material, and then transports that to a CP (delivery point). Between the pickup point and CP, the DV may visit an RS (if necessary). After the delivery, the DV returns to the DB. The authors propose four different models, the last of which determines the locations of both DBs and RSs, where the origin and destination DBs are not necessarily the same. The problem aims to minimize the total cost consisting of the fixed costs of both DBs and RSs and the transportation costs of DVs. A case study based on Brussels, Belgium is conducted.

One paper has considered DB and RS locations for UGVs. [Chalupa and Nielsen \(2018\)](#) present two formulations for a two-level facility location problem where DVs (autonomous ground vehicles) are used. Both models determine the locations of RSs (main stations) and DBs (storage locations-substations) and the assignment of demand nodes to DBs, and of DBs to RSs. Both models minimize the sum of the fixed costs of opening DBs and RSs and the total transportation cost. While the first model calculates the transportation cost based on the distance, the second one uses both distance and demand.

Finally, one paper has considered the location of both DB-Ts and RSs for drones in discrete space. [Cicek et al. \(2022\)](#) propose the communication-aware drone delivery problem with two communication quality constraints. One constraint limits the number of times a DV can change the node to which it communicates while flying on an arc. The second constraint limits the amount of time a DV can be traveling with poor-quality communications on an arc. DVs start and end their tours at DBs (depots), and can visit DB-T (other depots) to be resupplied, and visit RSs (charging stations) to be recharged. The problem aims to minimize the total flight distance while satisfying the handover, outage, battery, and time window constraints. The authors present a MILP formulation and a genetic algorithm.

5. Problems with no interaction between drones and other vehicles

This section analyzes the literature where both OV and DVs are used to make deliveries, but there is no interaction between these two types of vehicles, meaning they make deliveries independently. [Table 10](#) shows the studies in this section based on the same features used in [Table 9](#). [Table 10](#) indicates that all four studies consider a cost-based objective function with aerial DVs serving a single CP per trip without

time windows. Additionally, all problems except one ([Chowdhury et al., 2017](#)) are defined on a discrete space for a commercial application without considering uncertainty.

One of the first studies that consider location decisions in drone delivery problems is ([Chowdhury et al., 2017](#)), which proposes DVs as an alternative to OVs (trucks) to deliver relief items in the response phase after a disaster. The authors present a continuous approximation model that decides: (i) DB (distribution center) locations, from which DVs will be launched, (ii) service regions of DBs, and (iii) ordering quantities. The objective is to minimize the sum of fixed costs of opening DBs, ordering and holding costs of emergency supplies, and expected transportation costs of OVs and DVs. The total transportation cost considers the uncertainty of network resilience by assigning probabilities of using an OV (if roads are accessible) or a DV (if roads are not accessible). Additionally, DV transportation cost is calculated using a nonlinear function for energy consumption during the different stages of a DV flight, including climbing, hovering, descending, turning, acceleration and deceleration, rotation, and constant speed. The authors implement a two-phase continuous approximation approach where the first phase divides the entire region into sub-regions, and the second phase uses continuous approximation analysis in each subregion. Three coastal counties of Mississippi, USA are considered in the case study. [Baloch and Gzara \(2020\)](#) present a competitive facility location problem for an e-retailer offering same-day delivery services including a fast DV delivery. The authors consider a service-based competition, where different services offered by the e-retailer compete with each other, as well as with delivery services from other physical stores. A multinomial market share model is used to incorporate the competition. The objective function maximizes the total profit calculated by subtracting the delivery costs, fixed cost of DBs (facilities), and service costs from the total revenue. The authors present a MINLP formulation, which is solved by a logic-based Benders decomposition approach. A case study based on New York City is carried out.

Two studies in this section aim to locate RSs. [Cokyasar et al. \(2021\)](#) proposes a drone delivery network problem (DDNP), where one or more RS (automated battery swapping machines) can replace DV batteries. The problem determines: (i) whether to deliver by OV or DV, (ii) locations of RSs, (iii) number of DVs, and (iv) utilization rate at RS locations. The objective is to minimize the sum of fixed RS cost, fixed DV cost, DV and OV delivery cost and the cost associated with lead time due to battery swapping operations, including the waiting time in a queue. The problem considers possible congestion at an RS modeled as an M/D/1 queue. For DV deliveries, the routes for the forward journey to CPs (customers) and return journey to DBs (distribution centers) are optimized separately. The authors develop a MINLP to account for nonlinear waiting times of DVs at RSs. The authors propose an exact solution approach based on a cutting-plane method. [Cokyasar \(2021\)](#) study an e-commerce drone delivery problem, similar to the DDNP of [Cokyasar et al. \(2021\)](#), but with an additional battery inventory cost and a partitioning of demand at each location into DV-only, OV-only and mixed delivery modes. The authors present a MINLP, and reformulate it as a mixed-integer quadratically-constrained program by relaxing the queuing constraints, which are iteratively added to the program in a cutting plane method.

6. Problems where other vehicles resupply drones

This section reviews research where OVs resupply DVs, but DVs never ride on OVs. For drone delivery location problems with riding, see the companion paper ([Dukkanci et al., 2023](#)). The seven references in this section decide the locations of DBs and one of them ([Liu et al., 2021](#)) also decides the locations of OBs. [Table 11](#) indicates that all seven studies are defined on a discrete space for a commercial application where uncertainty is not considered. The literature considers cost (4) and time (4) related objective functions, and most problems are

Table 10

Classification of literature on facility location problems with no interaction between OV and DVs.

	Facility	Objective	Space	DV	# customers	Feature	App	Form.	Sol App E/H
Chowdhury et al. (2017)	DB, OB	C(a,c,d,e)	C	A	1	U	Hum		-/CA
Baloch and Gzara (2020)	DB, OB	C(a,c,e)	D	A	1		Com	MINLP	BD/-
Cokyasar et al. (2021)	RS	C(a,b,c,d,e)	D	A	1		Com	MINLP	CP/-
Cokyasar (2021)	RS	C(a,b,c,d,e)	D	A	1		Com	MINLP	CP/-

C(.,.) is a cost objective, T(.,.) is a time objective, V is a coverage objective and O is other objectives, where details are provided in Table 5.

Table 11

Classification of literature on facility location problems with OV resupplying DVs, where DVs never ride on OV.

	Facility	Objective	Space	DV	# customers	Feature	App	Form.	Sol App E/H
Schermer, Moeini, and Wendt (2019)	DB	C(a,c,e),T(g)	D	A	1		Com	MILP	CS/-
Poeting, Schaudt, and Clausen (2019)	DB	T(f)	D	G	1	TW	Com		-/SA
Bakach et al. (2021)	DB	C(c,e),O	D	G	1	TW	Com	MILP	-/SO
Liu et al. (2020)	DB	C(c,d,e)	D	G	>1		Com	MILP	-/GA
Alfandari et al. (2022)	DB	T(h)	D	G	1	TW	Com	MILP	CS,BBC/-
Kloster, Moeini, Vigo, and Wendt (2023)	DB	T(g)	D	A&G	1		Com	MILP	CS/Math,ILS
Liu et al. (2021)	OB, DB	C(a,b,c,d,e),O	D	G	>1	TW	Com	MILP	CS/AIA

C(.,.) is a cost objective, T(.,.) is a time objective, V is a coverage objective and O is other objectives, where details are provided in Table 5.

defined for ground drones (6) to serve a single customer per trip (5) with time window restrictions (4).

Schermer et al. (2019) propose the traveling salesman drone station location problem where an OV (truck) is routed between DBs (drone stations) to supply them, and then DVs make deliveries from DBs to CPs (customers). The authors present two variants of the problem where the first one minimizes the makespan, and the second one minimizes the sum of fixed costs of opening DBs and transportation costs of the OV and DVs. They present MILP formulations, and solve the first variant by an optimization solver.

Four studies in this section studied a similar two-echelon problem for ground drones. Poeting et al. (2019) present a delivery problem where (i) in the first echelon, an OV departs from an OB (hub) and delivers its packages to a set of DBs (micro-depots), and then (ii) in the second echelon, DVs (robots) depart from DBs to serve each assigned CP one-by-one. The problem aims to minimize the total distance traveled while determining the route of the OV and the allocation of CPs and DVs over several consecutive days. The problem considers the time windows for CPs. The authors present a simulation model including a GIS environment and simulated annealing heuristic. Bakach et al. (2021) study a similar problem, divided into two subproblems corresponding to each echelon. In the first subproblem, the optimal number of DBs is found first and then for the given number of DBs, the OV route is determined to minimize its travel cost. In the second subproblem, the operational cost of DVs is minimized to find the assignment of customers to DBs. The authors develop MILP formulations for both subproblems with and without time windows. Liu et al. (2020) consider a two-echelon vehicle routing problem where in the first echelon, OVs (vans) leave the OB (depot) to visit several CPs, some of which are used as DBs (satellites), from which DVs (autonomous delivery vehicles) depart to serve several CPs before returning to DBs. The problem determines the routes of OVs and DVs, assignment of CPs to DBs, and flow on each arc of the two-echelon network while minimizing the sum of transportation and emission costs of OVs and DVs, and the handling cost at DBs. The problem considers the capacity limitations of OVs, DVs and DBs. The authors develop a two-step clustering-based hybrid genetic algorithm and particle swarm optimization algorithm to solve the problem. Alfandari et al. (2022) focus on a similar problem proposed by Bakach et al. (2021) where an OV leaves an OB (depot) to deliver packages to DBs (local distribution facilities), and then DVs (autonomous robots) make deliveries from the DBs to CPs. The problem aims to minimize three different tardiness-based objectives: (i) maximum tardiness, (ii) total tardiness and (iii) number of late deliveries given deadlines for CPs. The authors develop a MILP formulation and a branch-and-Benders-cut method as an exact solution approach.

Kloster et al. (2023) is the only study that proposes to use both aerial and ground drones for deliveries in the multiple traveling salesman problem with drone stations. The proposed network consists of an OB (depot), DBs (packet stations) and CPs. Trucks leave the OB and visit either DBs to supply those points or CPs to make deliveries. The aerial and ground drones make deliveries to CPs directly from DBs. The objective is to minimize the makespan. The authors develop a MILP formulation and propose two variants of a decomposition-based matheuristic and a metaheuristic based on the iterated local search.

One study in this section aims to locate both DBs and OBs (not coincident) in a discrete space with ground drones. Liu et al. (2021) study a multi-objective two-echelon location routing problem where in the first echelon OVs (vans) depart from OBs (depots) and are routed between DBs (satellites), from which DVs (driverless delivery robots) depart to deliver to multiple CPs. Three objective functions are considered: (i) minimize the sum of fixed costs for OBs and DBs, fixed costs for OVs and DVs, transportation costs of OVs and DVs, and average handling costs at OBs and DBs, (ii) minimize the total emissions from operating OBs and DBs, transportation by OVs and DVs, and loading/unloading operations at OBs and DBs, and (iii) minimize a total violation penalty to increase customer satisfaction. The problem determines the locations of OBs and DBs, assignments of CPs to DBs, and DBs to OBs, routes of OVs and DVs while satisfying the capacity of OBs, DBs, OVs and DVs, time windows and a battery time limit. The authors present a MILP formulation and develop a hybrid artificial immune algorithm.

7. Problems where drones (re)supply other vehicles

This section reviews six references with DVs and OVs, where DVs (re)supply OVs at (re)supply points (SP) and OVs make deliveries. The locations of SPs are implicit decisions of the problem. Table 12 presents the features of these problems. Six papers study a discrete space with UAV deliveries for a commercial application. One paper allows DVs to make multiple deliveries in a flight and decides the locations of DB-Ts along with SPs. These models include cost (3), time (2) and coverage (1) related objectives.

Dayarian et al. (2020) introduce the VRP with drone resupply, where DVs regularly resupply OVs (delivery trucks). The resupply can take place at any location where the OV is stationary. The study mainly focuses on the single-DV, single-OV versions of the problem. The primary and secondary objective functions of the problems are to maximize the percentage of orders served and to minimize the total operational cost, including OV and DV transportation costs, respectively. The authors develop two-stage heuristic solution approaches.

Table 12

Classification of literature on facility location problems with DVs (re)supplying OV.

	Facility	Objective	Space	DV	# customers	Feature	App	Form.	Sol App E/H
Dayarian, Savelsbergh, and Clarke (2020)	SP	C(c,d),V	D	A	1		Com		-/PSH
Moshref-Javadi, Van Cauwenberghe, McCunney, and Hemmati (2023)	SP	T(f)	D	A	1		Com	IP	CS/-
Pina-Pardo et al. (2021)	SP	T(g)	D	A	1		Com	MILP	CS/MatH
Dienstknrecht, Boysen, and Briskorn (2022)	SP	C(c,d,e)	D	A	1		Com	MILP	CS/PSH
Liu, Liu, and Qi (2022)	SP	O	D	A	1	TW	Com	MILP	CS/GH
Luo et al. (2022)	SP, DB-T	C(b,c,d,e)	D	A	>1		Com	MILP	CS/ILS

C(.,.) is a cost objective, T(.,.) is a time objective, V is a coverage objective and O is other objectives, where details are provided in Table 5.

Moshref-Javadi et al. (2023) consider a similar problem where DVs carry packages from the DB (distribution center) to SPs (transshipment points) where OVs pick up the packages. First, an optimization model is solved to find the location of SPs to minimize total travel time for DVs and OVs. Then, the authors present a simulation model to compare the proposed system with truck-only delivery. Two case studies in an urban area (Boston) and a suburban area (Pittsfield) in Massachusetts, USA are conducted. This study is based on the thesis by McCunney and Cauwenberghe (2019). Pina-Pardo et al. (2021) introduce a new variant with package release dates. Unlike Dayarian et al. (2020) and Moshref-Javadi et al. (2023), orders (including package destinations) are not ready initially, so a DV is used to supply those items to the OV at an SP (meeting location), which is selected from a subset of the CPs. A multi-trip extension is presented where the OV makes multiple delivery routes and newly released orders can be picked up by the OV directly or resupplied by the DV. The authors develop MILP formulations and propose a matheuristic algorithm, where the problem is decomposed into subproblems for OV-routing decisions and for DV-resupply decisions. Dienstknrecht et al. (2022) study a similar problem in both static and dynamic settings. Different than the study of Dayarian et al. (2020), they consider the capacity of OVs and a cost-minimizing objective function consisting of the driving cost of the OV, tour cost associated with the driver wage, and flight cost of the DV. The problem determines the route of the OV and the scheduling of DV resupplies while synchronizing both vehicles. The authors develop a MILP formulation and heuristic solution approaches based on decomposition into subproblems for (i) routing OVs based on genetic algorithms and construction heuristics, and (ii) obtaining DV schedules by a dynamic program. Different from previous studies, Liu et al. (2022) consider drone resupply for multiple trucks and multiple drones to make on-time deliveries. The problem also assumes that truck routes are given in advance as it is based on an application where delivery sites such as residential complexes or office buildings receive multiple individual deliveries. The objective function maximizes the resupply payoff representing the total delivery value of packages resupplied by drones that can be delivered on time. The authors develop a time-expanded network flow problem and a greedy heuristic to solve the problem.

Luo et al. (2022) introduce a new pickup and delivery problem where multiple OVs equipped with a single DV that can make multiple visits in a single trip, and conduct pickup and delivery operations. The problem implicitly decides the locations of DB-Ts where OVs supply DVs, and the locations of SPs where DVs supply OVs by determining the routes of DVs and OVs. Routes must satisfy precedence constraints of requests, payload capacities of DVs and OVs, and range limitations of DVs. The problem aims to minimize the sum of duration costs of all deliveries and fixed costs of truck-drone pairs. The authors develop a MILP formulation and an iterated local search algorithm to solve the problem.

8. Problems with novel use of drones

This section presents an analysis of several novel applications involving facility location for drone delivery, which do not fit in other

categories. Table 13 shows the 14 studies in this section which consider locations of seven different facilities. Most of the studies have either cost-based (10) or time-based (3) objective functions, and are defined on discrete space (12). Different than the other categories, the majority of these studies consider ground DVs (8). While 4 studies consider time windows, the consideration of uncertainty (2) is quite limited.

Schwerdfeger and Boysen (2020) propose a dynamic location problem, in which mobile (autonomous) parcel lockers, which can be viewed as DVs, are relocated over a planning horizon so that customers are within a given distance of their assigned lockers. The objective is to minimize the locker fleet while ensuring customer satisfaction. The authors developed three different MILP formulations and a greedy heuristic algorithm. Reed et al. (2022b) introduce a human-DV delivery system for an urban environment where a DV (autonomous vehicle) can drop off a delivery person who then makes the delivery on foot. After delivery, the DV picks up the delivery person at another location and the person and DV travel together to the next drop-off. At the drop-off points (SPs), DVs resupply the delivery person as an OV. The objective minimizes the completion time of the delivery tour. The authors characterize the optimal solution on a rectangular grid of customers as a reasonable assumption for an urban area. Reed et al. (2022a) extends this model by considering a general geography of customers and allows the delivery person to wait for the DV. The authors develop an IP formulation along with some preprocessing rules and valid inequalities. The authors conduct a case study on nine counties in Illinois, US, where each county has a different degree of urbanization. Campbell et al. (2023) introduce the “multi-purpose K -drones general routing problem” where multiple drones conduct both deliveries to discrete points and sensing activities over continuous areas. The continuous area for sensing is modeled as a set of parallel lines, which must be traversed by the DV to cover (e.g., map) the area. The problem determines the DV routes, including entry and exit points of DVs on these lines, to minimize the total duration, while satisfying DV range and capacity limits. The authors discretize the lines by defining a set of entry and exit points for the DV and develop an IP formulation, along with a matheuristic algorithm and a branch-and-cut procedure.

Several studies have addressed problems to locate platooning points. Scherr et al. (2019) propose a service network design problem with DVs (autonomous vehicles) and OVs (manually operated vehicles). In some zones of the network DVs can operate alone, but in other zones DVs must be guided by OVs in platoons. The problem seeks to locate the PPs (platoon points) where DVs and OVs start and end platoons while minimizing the sum of transportation costs and fixed costs. The authors develop a MILP formulation, which decides the fleet mix and routing of the vehicles and goods. Scherr et al. (2020) extends this problem to allow outsourcing of deliveries to third-party service providers and not allowing split deliveries. The authors develop an algorithm based on the dynamic discretization discovery scheme. They propose valid inequalities and two enhanced versions of the algorithm, along with a heuristic solution approach. Scherr et al. (2022) further extends these platooning studies by considering a centralized tactical planning problem from the perspective of a platooning service provider, which owns OVs, to determine the fleet size and scheduling of platooning

Table 13
Classification of literature on facility location problems with novel use of DVs.

	Facility	Objective	Space	DV	# customers	Feature	App	Form.	Sol App E/H
Schwerdfeger and Boysen (2020)	DB-T	O	D	G	>1	TW	Com	MILP	CS/GH
Reed, Campbell, and Thomas (2022b)	SP	T(g)	D	G	1		Com	IP	CS/PSH
Reed, Campbell, and Thomas (2022a)	SP	T(g)	D	G	1		Com	IP	CS/-
Campbell, Corberán, Plana, Sanchis, and Segura (2023)	CV*	T(f)	C	A	>1		Hea	IP	BC/MatH
Scherr, Saavedra, Hewitt, and Mattfeld (2019)	PP	C(b,c,d)	D	G	>1		Com	MILP	CS/-
Scherr, Hewitt, Saavedra, and Mattfeld (2020)	PP	C(b,c,d)	D	G	>1		Com	IP	CS,DDD/PSH
Scherr, Hewitt, and Mattfeld (2022)	PP	C(b,c,d)	D	G	>1	U	Com	IP	CS/-
You, Miao, Zhang, and Xue (2020)	PP	C(b,c,d,e)	D	G	>1	TW	Com	MILP	CS/ACO
Xue, Lin, and You (2021)	PP	C(c,d,e)	D	G	>1	TW	Com	MILP	CS/TW
Peng and Xue (2023)	PP	C(c,d,e)	D	G	>1	TW	Com	MILP	CS/SA
Watanabe, Kenmochi, and Sasa (2021)	PP	C(c)	C	G	1		Com		-/CA
Enayati, Li, Campbell, and Pan (2023)	DB, RS, CP	C(a,c,d,e)	D	A	1		Hea	MILP	CS/-
Enayati, Campbell, and Li (2023)	DB, RS, CP	C(a,c,d,e)	D	A	1		Hea	MILP	CS/-
Wang, Jiang, and Qi (2023)	CP	C(a,b,e),O	D	A	1	U	Hea	MILP	CCG/-

C(.,.) is a cost objective, T(.,.) is a time objective, V is a coverage objective and O is other objectives, where details are provided in Table 5.
CV*: the coverage point where coverage by a drone begins or ends.

services. The problem aims to maximize the profit calculated by subtracting the costs of running the fleet from the revenue generated by the platooning service. As the problem considers the uncertainty of platooning demand, the authors present a two-stage integer programming formulation. Another area of problems that consider platoons with DV (autonomous trucks) is the local container drayage problem, which plays an important role in maritime transportation. You et al. (2020) introduce a generic model for platooning in a local container drayage problem. A semi-automated platoon consisting of an OV (a lead truck) with a driver and DVs (follower autonomous trucks) is considered. In this problem, the same customer can be served by two different drivers; so a truck can be part of two different platoons. The proposed problem determines the driver and truck routes (thus the locations of PPs) while minimizing the total cost including driver cost, truck deployment cost, and truck travel cost. The authors propose a MILP formulation and a heuristic algorithm with a learning mechanism based on ant colony optimization. Xue et al. (2021) extends the study of You et al. (2020) by considering the impact of fuel consumption savings on platooning. In this problem, a truck cannot be a part of two different platoons, and the cost-based objective consists of driver and fuel costs. The authors present a MILP formulation and a tabu search solution algorithm. Peng and Xue (2023) further extends the studies of You et al. (2020) and Xue et al. (2021) by focusing on the impact of empty and loaded trucks on the fuel consumption of the platoon. Therefore, the fuel cost component of the objective function is constructed in more detail compared to the one in Xue et al. (2021). The authors develop a MILP formulation and a simulated annealing algorithm. Watanabe et al. (2021) study a facility location problem for the formation of truck platooning for DVs (unmanned trucks). A continuous approximation method is used to compare two different optimal location strategies for platooning, along with a strategy with no platooning in terms of the average distance traveled.

The final area of novel applications involves deciding the locations of customer points, along with other facility types. Enayati, Li, Campbell, and Pan (2023) address multi-modal vaccine distribution using two types of DVs (drones) along with several types of OVs (e.g., trucks, boats and planes). The proposed problem determines the locations of CPs (local distribution centers), DBs (drone bases) and RSs (drone relay stations) and the vaccine flows from central depots to the CPs, while minimizing the sum of fixed facility costs, fixed network costs and variable transportation costs. The model includes drone range limits and a cold chain travel time limit for vaccines. The authors develop two mathematical models and solve a case study in Vanuatu. Enayati, Campbell, and Li (2023) extend this problem by including transfer times between transportation modes, setup times at the start of trips and the impact of payload weight on the DV transportation cost. Wang, Jiang, and Qi (2023) study a two-stage robust drone-based vaccine

distribution problem to address vaccine shortages. The model is a three-layered network where DVs carry vaccines from a DB to CPs (health facilities) and the people visit these CPs to get vaccinated. They determine the location of CPs, fleet size of DVs and vaccine scheduling in the first stage and then reschedule vaccines based on the availability in the second stage. The objective function includes social and economic benefits, including benefits from successful vaccinations, penalties for vaccination delays and wastage, and relevant costs for travel and inventories. The model is solved with two column-and-constraint generation algorithms for a case study in China.

9. Summary and future research directions

This survey analyzed 82 papers which appeared in 44 different journals, with some being conference papers or unpublished works (available online). Appendix B presents a detailed analysis of the number of papers per journal. An overall summary of the papers covered in this review, from Tables 8–12 follows:

- The most common objectives considered in this research are cost-based (54%), followed by time-based (35%), with about 16% of the studies considering a coverage-based or other objective function. Overall, 95% of the problems that include both DVs and OVs (Sections 5, 6, 7) aim to minimize either a cost-based or a time-based objective function, showing limited research with coverage-based or other objective functions.
- About 89% of the research is for discrete space problems, and 78% of all studies consider aerial drones (UAVs). In problems that consider drones alone with no other vehicles (Section 4), 96% of the research is for UAVs, which shows ground drone (UGV) applications are understudied.
- The large majority of the literature (73%) assumes that drones make a single delivery per trip and only 17% of the literature separately considers time window restrictions and uncertainty. None considers both at the same time.
- Over half of the research is based on commercial applications (59%), but both healthcare (27%) and humanitarian (15%) applications have been explored to some degree. In the literature that includes OVs (Sections 5, 6, 7), 95% of the research considers a commercial application, with only one paper (Chowdhury et al., 2017) having OVs and DVs in healthcare and humanitarian settings.
- Solution approaches used in the literature are fairly evenly divided among exact approaches (39%), heuristic approaches (32%) and both approaches (29%). Exact solution approaches are dominated by the use of a commercial solver (82%), with only 18% of the papers developing an exact solution approach such as branch-and-cut, Benders decomposition, etc.

We remark that many studies in this review did not focus on the unique features DVs, which makes the problems different from classical problems with trucks (or other vehicles). We encourage more work to consider realistic drone energy consumption models, as, for example, in [Bruni and Khodaparasti \(2022\)](#), [Bruni et al. \(2023\)](#), [Chauhan et al. \(2019\)](#), [Chowdhury et al. \(2017\)](#) and [Zhu et al. \(2022\)](#). We also note that only 15% of the studies locate more than one facility type, although real drone delivery systems often include several facility types.

9.1. Computational aspects

This subsection discusses some computational aspects for the studies reviewed in this survey. Our aim is not to compare the reviewed studies in terms of the size of the instances solved, as there are many factors that affect the instance size and the solution times including the nature of the demand and its geographic distribution, modeling assumptions, configurations of the computer used for computational experiments, programming language, commercial solver, etc. Here, we seek to provide a more general perspective on some computational aspects of this type of problem. We derive the following insights from the studies reviewed:

- For the studies discussed in Section 4 (delivery with only DVs), the problems with coverage-based objective functions (in particular AED delivery problems) have allowed the solution of larger instances compared to problems with cost and time-based objective functions. While for the latter type, instances up to 100 nodes can be solved, for the coverage-type problems, the instance size increases to over 10,000 nodes for some basic settings. For most coverage-type problems, commercial solvers are used, whereas problems with cost- and time-based objectives tend to be solved with exact approaches (e.g., [Park et al., 2022](#); [Xia et al., 2023](#); [Zhu et al., 2022](#)) or heuristics (e.g., [Ghelichi et al., 2022](#); [Golabi et al., 2017](#); [Kim et al., 2019](#); [Macias et al., 2020](#)). All studies reviewed in Section 5 that discuss problems with no interaction between DVs and OV have cost-based objective functions and develop exact solution approaches, rather than using commercial solvers. In contrast, researchers who studied problems with resupplying operations in Sections 6 and 7 generally focus on developing heuristic algorithms, as commercial solvers seem limited to solving only small instances (15 nodes). In general, more studies should focus on exact solution approaches to have alternative ways to find the optimal solution for larger-sized instances, especially for the settings with cost-based objective functions.
- The calculation of drone energy consumption to model cost or DV range is an important factor that adversely affects the size of the instances solved to the optimality ([Bruni & Khodaparasti, 2022](#); [Zhu et al., 2022](#)). There are several energy consumption models in the literature and the choice has an impact on the size of the problem that can be solved. As we discussed in Section 3, there are complex non-linear energy consumption models that complicate the process of finding optimal solutions. In that regard, [Zhu et al. \(2022\)](#) is an important study that shows the impact of considering a more realistic model for energy consumption on the solution process. The authors propose two exact solution approaches (column-and-constraint generation and Benders decomposition) for the proposed problem with a simplified version of linear energy consumption by [Figliozzi \(2017\)](#). On the other hand, they develop heuristic algorithms based on column-and-constraint generation for the problem with non-linear ([Liu et al., 2017](#)) and linear ([Figliozzi, 2017](#)) energy consumption models as finding the optimal solution is computationally difficult. To solve larger-sized instances, it seems likely that some judicious assumptions have to be made regarding the (linear or non-linear) energy consumption models, and this is the heart of good modeling. However, researchers should be cautious about using energy

and cost approximations as there are not yet good independent sources of energy consumption and cost data for large-scale drone delivery. Needless to say, optimal (or good) solutions for a poor model may be of limited practical value.

- Considering uncertainty generally increases computational difficulty, as in related delivery and location problems. However, the most important sources of uncertainty for real-world drone delivery applications are not well understood. For example, uncertainty from weather effects may be much more significant for small DVs than for large vehicles like trucks. There are also large uncertainties in the costs for DV operations and for drone facilities, and in DV recharging/battery swapping costs and times, given these are still relatively new, and technologies continue to advance.

9.2. Future research

In spite of the intensity of recent drone delivery research, there are still a large number of important research problems to address, especially as drone delivery becomes more widespread around the world. In the remainder of this section we discuss some future research directions that we believe are especially interesting or important.

One important direction is to locate more different types of facilities in one model to better reflect real-world options. [Table 3](#) indicates that although several studies consider locating both DBs and RSs together, research on locating other types of facilities is quite scarce. In particular, researchers might study problems involving the location of temporary drone bases and (re)supply and platooning points. It is also important to consider each facility with its own requirements (equipment or space) and cost based on the application. Additionally, research on drone delivery may address the problem of locating holding points (HPs), since these can be important in practice. Including holding points can require the integration of queueing models, and some studies already consider queuing at RSs ([Cokyasar, 2021](#); [Cokyasar et al., 2021](#)) or for the availability of drones ([Boutilier & Chan, 2022](#); [Wang, Zhao, Cheng, & Qi, 2023](#)). Going further, locating new types of facilities with different features would be interesting. One example is package depots (PDs), which indicate the origins of packages and may be distinct from the vehicle and drone depots. For example, with restaurant deliveries there might be several restaurants (PDs) from which drones launched at a DB could pick up one or more meals (packages) to deliver to a customer. Except for a few studies (e.g., see [Dhote & Limbourg, 2020](#); [Gómez-Lagos et al., 2022](#); [Pinto et al., 2020](#)), current models assume that packages originate at a DB or DB-T, or that the locations of package origins (PDs) are given. However, locating PDs can be appropriate in various applications where customer demands are served from more than one location.

More research would also be useful on problems where DVs and OV work together for healthcare and humanitarian applications. The rich set of models for commercial settings with DVs and OV may well extend to these other settings. Another interesting potential application of drone delivery is delivering non-tangible (non-physical) items such as software to be installed to a computer, energy to recharge sensors as in [Jin et al. \(2021\)](#), or downloading data (providing communication) as in the studies of [Cicek, Gultekin, and Tavli \(2019\)](#), [Cicek, Shen, Gultekin, and Tavli \(2021\)](#). Other extensions to different facility types are to further consider mobile facilities, including automated mobile facilities (e.g., mobile recharging points or mobile storage lockers as customers), and dynamic location problems. One example with mobile RS is ([Lorenz, Mimmo, Otto, & Vigo, 2022](#)). Delivery to moving CPs (customers) is also not unreasonable as for delivery to ships approaching port or delivery of packages to airships operating as mobile warehouses.

Exploring the tradeoffs of delivery frequency, DV size and inventory aspects is certainly an important research direction. This can address how demand should be aggregated over time and space, and what

is the optimal drone size (and payload) for a particular setting. As drone costs increase with the drone size and payload, more accurate modeling of drone costs based on measured drone performance in field experiments would be useful to provide realistic data. Existing inventory-location research could be extended to include drones with their unique characteristics. For instance, [Wu, Huang, Liang, Zhang, and Zhang \(2022\)](#) study the capacitated facility location and production planning, which could be extended by including dynamics of drone delivery.

As much drone delivery research to date has focused on rather narrow problems in idealized settings, an important future research direction is to assess the efficient use of drones in broader delivery ecosystems. Further work is needed on integrating drones with other vehicles besides trucks or transit vehicles, including unmanned aerial and ground vehicles (for one example, see [Kloster et al. \(2023\)](#)). This includes large heavy lift (e.g., 500 kg payload) drones and long-range drones. Some research suggests multiple drone types offer important synergies in certain settings ([Enayati, Li, Campbell, & Pan, 2023](#)). Drones for sea surface and underwater deliveries or other operations can be also considered for future research ([Savkin, Verma, & Anstee, 2022](#)). Another area to further investigate is the use of drones in multilayer (multimodal) transportation networks. [Archetti, Peirano, and Speranza \(2022\)](#) and [Crainic, Gendron, and Kazemzadeh \(2022\)](#) discuss multilayer and multimodal transportation networks with conventional manned vehicles. Research should further explore whether drones can be used in such networks for better utilization of the proposed systems. Research is also needed to address the use of multipurpose DVs, which may both make deliveries and perform other activities while on a flight (e.g., monitoring, surveillance or communication). Several of these areas may require new types of drone facilities or multipurpose facilities that serve several vehicle types.

Research would also benefit from more attention to integrating various practical aspects for drone operations into theoretical OR models. The recent study of in-facility mobile drones for e-commerce applications of JD.com shows the potential for integrating real-world conditions in a drone setting ([Qin et al., 2022](#)). While strong assumptions (e.g., on costs, travel time and travel networks) lead to tractable models, practical issues may hinder implementation of these results. Only a few papers have considered practical issues such as drone separation constraints to ensure safety, communication between drones and base stations or avoiding obstacles. Further, more accurate nonlinear cost and emissions functions for drones should be better explored, as drone operations are more complex than for ground vehicles due to the complicated physics of flight. As discussed in Section 3, different cost functions can be included in objective functions, with more complex and accurate models considering the impact of payload, speed, battery level, wind, elevation, flight direction (e.g., level, climbing, descending), etc. Using more realistic objective functions might result in different optimal locations of facilities. In general, future research should strive to better link the studied problems to real-life industry applications, whether in commercial, healthcare or humanitarian settings.

Better modeling is also important to assess environmental impacts of drone delivery. This requires accurate emissions data on real delivery drones in real settings, and comprehensive modeling of vehicle, fuel and facility life cycles. There are some studies that compare the energy consumption and CO₂ emissions of drone-based and ground-based delivery services (e.g., [Kirschstein, 2020](#), [Cokyasar et al., 2023](#)), but more comprehensive comparisons with optimized use of relevant alternatives including alternative fuel vehicles are important. This is especially true in the wake of the new EU carbon pricing framework ([Albert & Hopkins, 2023](#)). Better modeling the details of drone recharging also needs research, given the range of options for recharging/refueling; and OR and engineering research for electric vehicles can be instructive (e.g., see [Perumal, Lusby, & Larsen, 2022](#)). Location decisions can be incorporated into such problems along with routing and scheduling

decisions to efficiently charge a large fleet of drones. One recent example from electric vehicle research with intra-route recharging and en-route battery swapping is ([Raeesi & Zografos, 2022](#)) which combines routing, scheduling and RS location decisions. Additionally, better modeling of the use of 3D space for both travel and charging of aerial drones should be investigated. For small aerial drones, research should consider how the vertical dimension can be exploited to improve operations (e.g., using multi-level DB and RS facilities).

Future studies should also address further details of drone flight operations such as takeoff, landing and hovering, which are widely ignored in the current OR literature, but they are quite important in practice. Using different types of drone models results in different requirements for takeoff and landing operations. For instance, while rotary-wing aerial drones do not require too much space for takeoff and landing, fixed-wing drones need additional space for those operations. Research that better models weather impacts on drones would also be very useful. Another important practical issue needed to be addressed is the integration of drone delivery service into air traffic management to ensure safe flights of these vehicles. In addition to the coordination among drones, the coordination between drones and other aerial vehicles such as planes or helicopters should be managed to ensure safe separation distances and to avoid collisions. This can result in delays and detours in drone travel that are generally not reflected in OR modeling for drone delivery ([Liu, 2021](#)). One pioneering research that focuses on this integration is ([Levin & Rey, 2023](#)), which studies the planning of drone delivery service in urban airspace. The same type of problem might arise for ground traffic management if there is a large number of ground drones. Integrating (aerial and/or ground) drone delivery services into traffic management might lead to more nonlinear modeling and different locations of drone facilities. Because drone delivery travel paths depend on the locations of the drone delivery facilities, optimizing the locations of such facilities should be done in light of the future drone travel paths and needs for collision avoidance and air traffic management.

Another future research area is to consider how optimization modeling of outdoor drone delivery problems might be applied to analogous problems inside facilities, such as warehouses. Different from the automated guided vehicle (AGV) system, in-facility autonomous mobile robots (drones) can communicate with other resources in a decentralized system. See for example, [Fragapane et al. \(2021\)](#) for a comprehensive survey. As seen in Table 3, the existing research considers location decisions in either discrete and continuous space. Drone delivery models that combine location decisions in both spaces should be further investigated, and researchers could benefit from the study of [Blanco, Gázquez, and Saldanha-da Gama \(2023\)](#) as a good starting point.

Finally, further research on drone delivery can leverage state-of-the-art modeling and solution approaches from non-drone problems as the drone-based research has not fully utilized the findings of non-drone-related studies. This includes areas like stochastic, robust or dynamic optimization, and using heterogeneous fleets and split deliveries. We also anticipate that advances in drone delivery research might lead to improved modeling and solutions for related non-drone problems. As the summary above showed, more research should also address problems with coverage and service objectives. As problem size and scope grows there will likely be a need for more research on tailored exact solution approaches. Certainly, better modeling is useful to solve larger problems, and along with classical exact and heuristic approaches, more studies might explore machine learning or reinforcement learning techniques. [Bengio, Lodi, and Prouvost \(2021\)](#) and [Kraus, Feuerriegel, and Oztekin \(2020\)](#) discuss how machine learning and deep learning techniques can be used to solve OR problems, respectively. For instance, [Boute, Gijbrecchts, Van Jaarsveld, and Vanvuchelen \(2022\)](#) present a roadmap to use deep reinforcement learning for inventory control problems. These studies could be good starting points to apply these learning techniques to drone delivery problems.

10. Conclusion

This study presents a comprehensive literature review of papers that address facility location problems for drone delivery, where drones do not ride on other vehicles. We first define the functions of drones and eight facility types to be located in drone delivery problems. Some of these facilities exist in problems where drones are the only vehicles used, while other facilities are needed for problems when drones operate with other vehicles in the delivery process. We discuss the unique features of drones relative to vehicles in classical problems, and propose a general methodological framework to identify the main elements in a facility location problem with drones, and illustrate these elements in various formulations. The main part of this paper is guided by a categorization of the literature based on the involvement of drones and other vehicles, and their interactions. The comprehensive review of the relevant literature includes tables that summarize the papers in terms of relevant dimensions. Finally, we present many promising directions for future research.

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Supplementary material

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